

72nd BPSC PRELIMS

प्रहार

Test Series

Test- 25/09

FULL LENGTH TEST

Solution

English



BPSC TEST-25/9 (SOLUTION)

Q 1. Ans. B

Explanation:

- As per the Census of India 2011, the **overall literacy rate of Bihar is 61.80%**, making it the state with the lowest literacy rate in the country at the time of the 2011 Census. The literacy rate is calculated for the population aged **7 years and above**. Bihar recorded a **male literacy rate of 71.20%** and a **female literacy rate of 51.50%**, indicating a significant gender gap in literacy. Therefore, **61.80%** is the correct overall literacy rate of Bihar as per the Census 2011.

Q 2. Ans. D

Explanation:

- Statement 1 is correct:** Among Bihar's districts in the Census 2011, Gopalganj recorded the highest sex ratio (about 1,021 females per 1,000 males, one of the few districts exceeding 1,000, owing to high male out-migration), while Munger recorded the lowest (about 876).
- Statement 2 is correct:** Bihar's rural sex ratio (about 921) was higher than its urban sex ratio (about 895) in 2011. This is largely because male-dominated migration to towns and cities lowers the urban ratio, while rural areas retain a relatively higher proportion of women.
- Statement 3 is correct:** Bihar's overall sex ratio was 918, while its child sex ratio (0 to 6 years) was 935. Since 935 is higher than 918, the child sex ratio exceeded the overall sex ratio. This inverts the all-India pattern (where the child sex ratio of 919 was lower than the overall 943), which is the sharpest trap in the set.

Q 3. Ans. A

Explanation:

- Professor Sachin Chaturvedi** is the Vice-Chancellor of **Nalanda University**. An eminent economist and policy researcher, he is leading the university's academic expansion and strengthening its global engagement from the Rajgir campus in Bihar. Prior to assuming this role, he served as the **Director General of the Research and Information System for Developing Countries (RIS), New Delhi**. Widely recognised as an institution builder, Professor Chaturvedi has made significant contributions to research on international development, trade, innovation, and public policy.

Q 4. Ans. C

Explanation:

- According to the **Bihar Budget 2026–27**, the **estimated fiscal deficit** for the financial year 2026–27 is **₹39,112 crore**, which is targeted at **3% of the Gross State Domestic Product (GSDP)** in line with the fiscal responsibility framework. The budget projects this level of fiscal deficit while maintaining a small revenue surplus and increasing capital expenditure for infrastructure and development. Therefore, **₹39,112 crore** is the correct estimate.

Q 5. Ans. A

Explanation:

- According to the Bihar Budget 2026–27, the state is expected to maintain a **revenue surplus of ₹1,143 crore**, equivalent to about **0.1% of GSDP**. A revenue surplus means that the state's revenue receipts exceed its revenue expenditure.

Q 6. Ans. C

Explanation:

- Statement 1 is correct:** According to the **Bihar Budget 2026–27**, the **Education, Sports, Arts & Culture** sector has been allocated approximately **₹70,141 crore**, making it the largest sectoral allocation in the budget. In comparison, the **Rural Development** sector has been allocated approximately **₹30,387 crore**. Hence, the allocation for the education sector is significantly higher than that for rural development.
- Statement 2 is correct:** The **Health** sector has been allocated approximately **₹20,230 crore** in the **Bihar Budget 2026–27**, which is considerably lower than the **₹70,141 crore** allocated to the **Education, Sports, Arts & Culture** sector. Therefore, the education sector allocation is also higher than the health sector allocation.

Q 7. Ans. A

Explanation:

- Statement 1 is correct:** The total electorate for the Bihar Assembly Election stood at approximately **7.45 Crores (7,45,26,858 eligible voters)**. This registered pool consisted of roughly **3.94 Crore male electors (3,93,79,366)** and **3.51 Crore female electors (3,51,45,791)**.

- **Statement 2 is correct:** Out of this registered pool, approximately 14 lakh first-time young voters were eligible to cast their ballots.
- **Statement 3 is incorrect:** While the election did record the highest voter turnout in the state's history since 1951, the overall turnout was around 67.13%.
- **Statement 4 is incorrect:** The female voter turnout registered at 71.78%, compared to a male turnout of 62.98%. However, this means female turnout exceeded male turnout by approximately 8.8%, not 12.80%

Q 8. Ans. B

Explanation:

- **Saurath Sabha**, popularly known as the “**biggest marriage market**”, is held at **Saurath village in Madhubani district**. It is a unique centuries-old tradition of the **Maithil Brahmin community**, where eligible brides' and grooms' families meet to arrange marriages. The gathering is conducted in accordance with the **Panji Prabandh** (genealogical record system) to ensure that marriages do not violate traditional lineage rules.

Q 9. Ans. D

Explanation:

- The **Praja Socialist Party (PSP)** was founded in **September 1952** through the merger of the **Socialist Party** and the **Kisan Mazdoor Praja Party (KMPP)**. The merger brought together prominent socialist leaders, including **Jayaprakash Narayan, J. B. Kripalani**, and **Acharya Narendra Deva**, with the objective of creating a united socialist political force in independent India. While the **Socialist Party** was led by leaders such as **Jayaprakash Narayan** and **Acharya Narendra Deva**, the **Kisan Mazdoor Praja Party** was headed by **J. B. Kripalani**. The formation of the PSP marked an important development in the evolution of India's socialist movement during the early post-independence period.

Q 10. Ans. C

Explanation:

- **Assertion (A) is true:** According to the **Köppen Climate Classification**, Bihar has a **Cwa type of climate**, which is described as a **humid subtropical climate with dry winters and hot summers**. The state experiences three distinct seasons—summer, southwest monsoon, and winter—with most of the annual rainfall occurring during the monsoon months (June to September).

- **Reason (R) is false:** Bihar does not have a **humid tropical rainforest (Af) climate**. A tropical rainforest climate is marked by **high temperatures and heavy rainfall throughout the year without a distinct dry season**, which is not the case in Bihar. Instead, Bihar has a **humid subtropical (Cwa) climate** with a **well-defined dry winter season**.

Q 11. Ans. B

Explanation:

- According to the **Indian Institutional Ranking Framework (IIRF) 2025**, **Bihar Sports University, Rajgir**, was placed at **4th rank** among **Government Sports Universities/Institutions** in India. The university achieved this recognition soon after receiving **UGC recognition under Section 2(f)**, reflecting its rapid progress in sports education and infrastructure.

Q 12. Ans. D

Explanation:

- The **Election Commission of India (ECI)**, through the **Office of the Chief Electoral Officer, Bihar**, designated **Nitu Chandra** and **Kranti Prakash Jha** as **State SVEEP (Systematic Voters' Education and Electoral Participation) Icons** for Bihar ahead of the **Bihar Assembly Elections 2025**. They were appointed to promote voter awareness, encourage electoral participation, and motivate citizens, especially first-time and young voters, to exercise their franchise through various outreach campaigns and public awareness programmes.

Q 13. Ans. C

Explanation:

- According to the **India State of Forest Report (ISFR) 2023**, Bihar is classified as a state with **low forest cover**. The state has a **forest cover of about 7,532.45 km² 8% of its geographical area**, which is significantly lower than the **National Forest Policy** goal of **33% forest and tree cover**. Most of Bihar's forests are concentrated in the southern districts, particularly in the **Kaimur, Rohtas, Jamui, Banka, Nawada, Gaya, and Munger** regions.

Q 14. Ans. A

Explanation:

- **Assertion (A) is true:** During the **retreating (withdrawal) phase of the southwest monsoon**, generally in **October and early November**, Bihar occasionally receives rainfall from **tropical cyclonic**

disturbances originating over the Bay of Bengal. Although these systems weaken before reaching Bihar, the rainfall they bring plays a significant role in the state's agricultural cycle.

- **Reason (R) is true:** The rainfall associated with these tropical cyclones is beneficial because it **helps the standing paddy crop attain proper maturity and replenishes soil moisture**, creating favourable conditions for the timely sowing of **rabi crops** such as wheat, gram, lentil, and mustard. This is the primary reason why such cyclonic rainfall is considered agriculturally important in Bihar.
- Since the **agricultural importance** mentioned in the Assertion is **directly due to** the benefits described in the Reason, the **Reason correctly explains the Assertion**.

Q 15. Ans. D

Explanation:

- According to the **Department of Mines & Geology, Government of Bihar**, **Rohtas district** is one of the mineral-rich districts of the state. It is known for the occurrence of **limestone**, lead-zinc, **bauxite**, and **pyrites**, which are of significant economic importance. Rohtas district possesses extensive **limestone** deposits, particularly in the **Kaimur plateau region**. Limestone is the principal raw material for the cement industry, and Rohtas is one of the major limestone-producing districts of Bihar. Rohtas district contains **pyrites**, an important sulphide mineral used in the manufacture of sulphuric acid and fertilisers.

Q 16. Ans. C

Explanation:

- The Revolt of 1857 in Bihar began with the **Danapur Cantonment uprising on 25 July 1857**. The rebel sepoys marched to **Arrah**, where they joined **Veer Kunwar Singh**, who assumed command of the rebels. His forces occupied **Arrah** two days later and laid siege to **Mr. Boyle's house**, where British officials had taken shelter.

Q 17. Ans. A

Explanation:

- **Babu Veer Kunwar Singh (1777–1858)** was one of the greatest heroes of the **First War of Independence (1857–58)**. The **zamindar of Jagdishpur** (present-day **Bhojpur district**, Bihar), he became the ruler of the estate in 1826 after the death of his father, **Raja Sahabzada Singh**.

- In **April 1858**, **Kunwar Singh** returned to **Jagdishpur** and achieved a decisive victory over the British forces led by **Captain Le Grand** on **23 April 1858**. During the campaign, after being wounded by a bullet in his wrist while crossing the **Ganga**, he is believed to have **severed his injured hand and offered it to the river**. **Veer Kunwar Singh** passed away on **26 April 1858**, leaving behind a legacy of courage and patriotism.

Q 18. Ans. C

Explanation:

- **Statement 1 is incorrect:** The **Gandak River** enters India at **Valmikinagar** in **West Champaran district**, **not East Champaran**. Valmikinagar, located on the Indo–Nepal border, is the point where the river enters Bihar.
- **Statement 2 is correct:** The **Gandak River** enters India from Nepal at **Triveni**, near **Valmikinagar** on the Indo–Nepal border. The **Valmikinagar (Triveni) Barrage** has been constructed at this location for irrigation and flood management.
- **Statement 3 is incorrect:** After entering India, the **Gandak River** forms part of the boundary between **Bihar and Uttar Pradesh** and flows through the **Maharajganj and Kushinagar districts of Uttar Pradesh** before re-entering Bihar and eventually joining the **Ganga near Hajipur (Vaishali district)**.

Q 19. Ans. A

Explanation:

- **Assertion (A) is true:** Bihar is divided into **three agro-climatic zones**. **Zone I (North-West Alluvial Plain)** and **Zone II (North-East Alluvial Plain)** experience frequent flooding due to the Himalayan rivers such as the **Gandak, Burhi Gandak, Bagmati, Kamla-Balan, Kosi, and Mahananda**. In contrast, **Zone III (South Bihar Alluvial Plain)** is comparatively **drought-prone**, as it depends more on rainfall and rivers originating from the peninsular plateau, which have relatively lower and more seasonal flows.
- **Reason (R) is true:** The **River Ganga** divides Bihar into two major physiographic regions. **Agro-climatic Zones I and II** are situated **north of the Ganga**, where numerous Himalayan rivers frequently cause floods. **Zone III** lies **south of the Ganga**, where rainfall variability and the absence of perennial Himalayan rivers make drought a more common concern.
- The **location of the agro-climatic zones relative to**

the Ganga and the river systems is the fundamental reason why Zones I and II are flood-prone, while Zone III is comparatively drought-prone. Hence, the Reason correctly explains the Assertion.

Q 20. Ans. C

Explanation:

- According to Bihar Government publications on the Buddhist Circuit, the West Champaran region is associated with the tradition that Prince Siddhartha renounced his royal robes and embraced the life of a monk before beginning his quest for enlightenment. This association makes West Champaran an important destination on Bihar's Buddhist Circuit.

Q 21. Ans. B

Explanation:

- Kot Diji Ware is an important Early Harappan ceramic tradition, characterised by thick, sturdy wheel-made red ware decorated with black-painted horizontal bands, wavy lines, loops, and simple geometric motifs. Named after the site of Kot Diji in Sindh (Pakistan), this pottery represents the transitional phase between the regional Chalcolithic cultures and the urban Mature Harappan Civilisation.
- **Additional Facts:**
 - ✓ Hakra Ware is associated with the Hakra (Ghaggar-Hakra) culture and represents one of the earliest ceramic traditions of the northwestern subcontinent, predating the Early Harappan urban phase.
 - ✓ Sothi-Siswal Ware belongs to the Sothi-Siswal cultural complex of Rajasthan, Haryana, and adjoining regions, and is distinguished by its own characteristic painted pottery traditions.
 - ✓ Amri Ware is associated with the Amri culture of Sindh and is generally characterised by comparatively finer pottery with painted geometric and naturalistic designs rather than the thick, sturdy red ware typical of Kot Diji.

Q 22. Ans. C

Explanation:

- The Ashokan Pillar Edicts are royal proclamations issued by Emperor Ashoka to communicate his policy of Dhamma and principles of ethical governance. Engraved on polished sandstone pillars erected across the Mauryan Empire, these inscriptions deal with subjects such as non-violence, religious tolerance, welfare of subjects, justice,

and administrative reforms. They were intended to convey the emperor's directives directly to the people.

● **Additional Facts:**

- ✓ Prashastis are eulogistic inscriptions composed in praise of kings and their achievements. A famous example is the Allahabad Pillar Prashasti of Samudragupta, composed by Harishena.
- ✓ Land grants record the donation of land by rulers or local authorities, often to Brahmanas, temples, monasteries, or educational institutions. Such inscriptions became particularly common during the Gupta and post-Gupta periods.
- ✓ Votive inscriptions commemorate religious dedications made by devotees, recording the donation or installation of stupas, images, temples, caves, or other sacred objects.

Q 23. Ans. B

Explanation:

- **Statement 1 is incorrect:** Painted Grey Ware (PGW) is not technologically superior to Northern Black Polished Ware (NBPW). PGW is a fine grey pottery decorated with black-painted geometric designs and is associated with the later Vedic period (c. 1200–600 BCE). In contrast, NBPW represents a more advanced ceramic tradition, distinguished by its highly lustrous, mirror-like black or steel-grey surface, achieved through sophisticated firing techniques.
- **Statement 2 is correct:** Northern Black Polished Ware (NBPW) is closely associated with the Second Urbanisation (c. 600–300 BCE) in the middle Gangetic plains. It is found at major urban centres such as Pataliputra, Kaushambi, Rajgir, Vaishali, and Ujjain, reflecting the growth of cities, trade, coinage, and the rise of the Mahajanapadas and the Mauryan Empire.

Q 24. Ans. A

Explanation:

- **Statement 1 is correct:** The Tripartite Struggle was fought among the Gurjara-Pratiharas, Palas, and Rashtrakutas for the control of Kannauj, which occupied a strategic position in the Ganga-Yamuna Doab. The region was economically prosperous, agriculturally fertile, and controlled important trade routes, making it a symbol of political supremacy in northern India.

- **Statement 2 is incorrect:** Although the Rashtrakutas, under rulers such as Dhruva and Govinda III, successfully invaded north India and defeated both the Gurjara-Pratiharas and the Palas, they did not establish long-term territorial control over Kannauj. Their campaigns were largely military expeditions aimed at demonstrating supremacy, after which they returned to the Deccan.

Q 25. Ans. B

Explanation:

- **Assertion (A) is true:** The Rig Veda shares profound linguistic, phonetic, and mythological similarities with the Zend-Avesta, the foundational religious text of ancient Zoroastrianism in Iran. The languages (Vedic Sanskrit and Avestan) are so closely related that entire verses can be translated from one to the other by simply applying standard phonetic shifts (e.g., the Vedic 's' often becomes an Avestan 'h', transforming the Vedic drink Soma into the Avestan Haoma, and the river Sarasvati into Harahvati).
- **Reason (R) is true:** Discovered in modern-day Turkey (Anatolia), the Boghazkoi inscriptions date back to roughly 1380 BCE. They document a peace treaty between two ancient empires (the Hittites and the Mitanni) and explicitly invoke four Vedic gods, **Mitra, Varuna, Indra, and the Nasatyas (Ashvins)**, as witnesses to the pact.
- **Why (R) does not explain (A):** While both statements are undeniable facts that support the broader historical theory of Indo-Aryan migrations, the discovery of a treaty in Turkey (R) is not the reason why Indian and Iranian texts share linguistic elements (A). The linguistic connection exists because both texts evolved from a common ancestral group (the proto-Indo-Iranians) before they split geographically. Therefore, they are two independent, parallel pieces of historical evidence.

Q 26. Ans. D

Explanation:

- **Vinaya Pitaka** → Monastic rules and regulations for Buddhist monks (2). The Vinaya Pitaka is one of the three baskets (Tipitaka) of Buddhist canonical literature, containing the rules of discipline and conduct for the monastic community (the Sangha).
- **Jataka Tales** → Stories detailing previous births of the Buddha (4). The Jatakas are a collection of stories narrating the previous births (both human and animal) of the Buddha as a Bodhisattva, illustrating moral and ethical lessons.

- **Angas** → Canonical texts written in Ardhamagadhi Prakrit (1). The Angas are the core canonical texts of Jainism (not Buddhism), composed in Ardhamagadhi Prakrit, forming part of the Jain Agama literature (traditionally twelve Angas, associated with the Shvetambara tradition).
- **Kathavatthu** → Philosophical interpretations of Theravada Buddhism (3). The Kathavatthu ("Points of Controversy") is one of the seven books of the Abhidhamma Pitaka, attributed to Moggaliputta Tissa and associated with the Third Buddhist Council (held at Pataliputra under Ashoka). It deals with points of doctrine and refutes heterodox views, representing the Theravada philosophical/doctrinal position.

Q 27. Ans. C

Explanation:

- **Pair 1 is correctly matched:** Megasthenes failed to understand the complex, birth-based Varna and Jati system. Instead, he observed the division of labour and incorrectly categorised Indian society into seven distinct occupational classes: Philosophers (Brahmins/Monks), Farmers, Herdsmen/Hunters, Artisans/Traders, Soldiers, Overseers (Spies), and Councillors/Assessors.
- **Pair 2 is correctly matched:** He provided a detailed account of the municipal administration of the Mauryan capital, Pataliputra. He noted that the city was governed by a central council of 30 members, which was divided into six specialised boards of five members each (handling matters like foreign residents, trade, vital statistics, and taxation).
- **Pair 3 is correctly matched:** Megasthenes famously (and erroneously) declared that slavery did not exist in India. Historians conclude he was likely blinded by his Greek perspective—Indian slavery (Dasa system), which allowed slaves to own property and buy their freedom, was far milder than the brutal chattel slavery of the Greco-Roman world. The contemporary Arthashastra by Kautilya directly contradicts Megasthenes by dedicating an entire chapter to the laws and rights governing slaves.

Q 28. Ans. A

Explanation:

- **Mehrgarh**, located in present-day Balochistan (Pakistan), is recognised as one of the earliest pre-Harappan (Neolithic-Chalcolithic) settlements in the Indian subcontinent. Occupied from around 7000 BCE, it provides the earliest evidence of

agriculture, animal domestication, mud-brick houses, pottery, craft production, and dentistry. Mehrgarh is widely regarded as the precursor to the later Indus Valley (Harappan) Civilisation.

● **Additional Facts:**

- ✓ Lothal is a Mature Harappan site in Gujarat, renowned for its dockyard, bead-making industry, and maritime trade.
- ✓ Surkotada, also a Mature Harappan site in Gujarat, is famous for providing evidence of horse remains, though their identification has been debated.
- ✓ Dholavira, situated in Gujarat's Rann of Kachchh, is one of the largest Harappan cities, noted for its sophisticated water management system, unique town planning, and the discovery of the Harappan signboard.

Q 29. Ans. C

Explanation:

- The Abhidhamma Pitaka is the third division of the Tipitaka (Tripitaka) and contains the systematic philosophical and psychological exposition of Theravada Buddhist doctrine. Unlike the Sutta Pitaka, which records the Buddha's discourses, the Abhidhamma analyses reality into its fundamental constituents (dhammas) and elaborates on concepts such as consciousness, mental states, causation, ethics, and the path to liberation. It is regarded as the highest and most analytical expression of Theravada Buddhist philosophy.
- **Additional Facts:**
 - ✓ Sutta Pitaka contains the discourses (suttas) of the Buddha and some of his principal disciples. It is divided into five Nikayas and forms the primary source of Buddhist teachings.
 - ✓ Vinaya Pitaka lays down the monastic rules and disciplinary code governing the conduct of Buddhist monks (bhikkhus) and nuns (bhikkhunis).
 - ✓ Dipavamsa is not a part of the Tripitaka. It is the earliest historical chronicle of Sri Lanka, composed in Pali, and provides valuable information on the spread of Buddhism in the island.

Q 30. Ans. A

Explanation:

- **Statement 1 is correct:** The **Nayankara system** was the backbone of provincial and military administration in the Vijayanagara Empire. Top military commanders (Nayakas) were granted

territorial land assignments known as **Amaram**. In exchange for these lands, the Amaranayakas were required to maintain a stipulated number of troops (infantry, cavalry, elephants) for the King and pay a fixed annual financial tribute to the central treasury.

- **Statement 2 is correct:** The **Ayagar system** was the definitive feature of village-level local administration. Each village was treated as an independent economic and administrative unit. Its affairs were managed by a council of **12 hereditary functionaries** collectively known as the Ayagars (which included vital roles like the Karnam/Senabhova [accountant], Gauda/Reddy [headman], and Talari [watchman]).
- **Statement 3 is incorrect:** Although the Ayagar offices were hereditary and they were compensated with tax-free land grants (Manyams or Inams), they could not freely sell or transfer their offices independent of the state. While historical records show that Ayagars could theoretically sell or mortgage their offices, such transfers of state-mandated administrative duties and tax-free lands strictly required the **permission and sanction of the government**.

Q 31. Ans. D

Explanation:

- **Statement 1 is correct:** The **Ur** was the most common type of assembly. It functioned in ordinary, non-Brahmanical villages (secular villages) and consisted of all the tax-paying landowners of that village.
- **Statement 2 is correct:** The **Sabha** (or Mahasabha) was an exclusive, highly organised assembly found only in Brahmadeya villages (tax-free lands granted entirely to Brahmins). Its membership was strictly restricted to Brahmin landowners.
- **Statement 3 is correct:** The **Nagaram** was an assembly specifically for towns, commercial centres, and market hubs. It was dominated by merchants, traders, and artisans, acting as an administrative and corporate body for commercial interests.
- **Statement 4 is correct:** The famous **Uttaramerur inscriptions** (issued during the reign of the Chola king Parantaka I) provide a remarkably detailed constitution for the Sabha. They explicitly laid down strict qualifications for candidates, including owning a certain amount of land, having a house built on legally owned land, falling within the age bracket of 35 to 70 years, and possessing deep educational qualifications (specifically, knowledge of the Vedas and administrative matters).

Q 32. Ans. A

Explanation:

- **Statement 1 is correct:** The Swarajists, led by C.R. Das and Motilal Nehru, favoured entering the legislative councils established under the Government of India Act, 1919. Their objective was to obstruct, expose, and ultimately weaken colonial rule from within by opposing government measures and making the councils unworkable.
- **Statement 2 is correct:** The No-Changers, led by Mahatma Gandhi, Rajendra Prasad, Vallabhbhai Patel, C. Rajagopalachari, and others, opposed council entry. They believed that national strength should be built through constructive work, including the promotion of khadi, village industries, national education, removal of untouchability, and Hindu-Muslim unity.
- **Statement 3 is incorrect:** C.R. Das and Motilal Nehru were the founders and principal leaders of the Swaraj Party, not the No-Changer group.
- **Additional Facts:**
 - ✓ The Swaraj Party was founded in 1923 by C.R. Das and Motilal Nehru after the suspension of the Non-Cooperation Movement.
 - ✓ At the Belgaum Session of the Indian National Congress (1924), presided over by Mahatma Gandhi, a compromise was reached allowing both the Swarajists and the No-Changers to function within the Congress.

Q 33. Ans. B

Explanation:

- **Assertion (A) is true:** Gandhi began a fast unto death at Yerwada Central Jail in September 1932, in protest against the Communal Award announced by Prime Minister Ramsay MacDonald, which granted separate electorates to the Depressed Classes along with other minority communities. Gandhi viewed separate electorates for the Depressed Classes as a device that would permanently divide Hindu society, and he undertook the fast to pressure political leaders into finding an alternative arrangement.
- **Reason (R) is true:** The Poona Pact, signed on 24 September 1932 between Gandhi (representing caste Hindus) and B.R. Ambedkar (representing the Depressed Classes), did indeed abandon the separate electorate scheme of the Communal Award. In its place, it introduced a system of reserved seats for the Depressed Classes within

joint (general) electorates, along with an increase in the total number of reserved seats compared to the original Award.

- **Why (R) is not the correct explanation of (A):**
- Although both statements are factually correct, the Reason describes an event that came after and as a result of the fast, not something that explains why the fast took place.
- **The correct causal relationship runs in the opposite direction:** Gandhi's fast (A) was the pressure that produced the Poona Pact (R), rather than the Poona Pact being the cause or explanation of the fast.

Q 34. Ans. C

Explanation:

- **First Anglo-Mysore War → Treaty of Madras, 1769:** The war ended in a stalemate, with the Treaty of Madras providing for mutual restitution of conquered territory and a defensive alliance between Haidar Ali and the English.
- **First Anglo-Maratha War → Treaty of Salbai, 1782:** This war, arising from English interference in Maratha succession disputes, ended inconclusively with the Treaty of Salbai, restoring the pre-war territorial status quo and securing about two decades of peace between the English and the Marathas.
- **Fourth Anglo-Mysore War → Death of Tipu Sultan, 1799:** The final war between the English and Mysore ended with the storming of Srirangapatna, in which Tipu Sultan was killed defending his capital, and Mysore's independent power was permanently broken.
- **Second Anglo-Maratha War → Treaty of Rajghat:** The Treaty of Rajghat (December 1805) was signed between the British East India Company and Yashwantrao Holkar, the Maratha chief of Indore, bringing the Holkar phase of the Second Anglo-Maratha War to a close. The British restored a substantial part of Holkar's territories under its terms.

Q 35. Ans. B

Explanation:

- **Jyotirao Phule:** He founded the **Satyashodhak Samaj** (Truth Seekers' Society) in Pune in 1873. Its primary goal was to secure social justice and equal rights for marginalised communities (specifically the Malis, Telis, Kunbis, and Dhangars) and to protect them from exploitation by Brahmins.
- **E.V.R. Periyar:** He launched the **Self-Respect Movement** in 1925 in Tamil Nadu. The movement aimed to dismantle Brahmanical hegemony, advocating for a rationalist society where backward

castes could live with dignity and equal human rights without the need for Brahman priests in marriages and rituals.

- **Narayana Guru:** In a revolutionary act in 1888, this Ezhava social reformer consecrated a Shiva idol at **Aravippuram** in Kerala. By doing so, he shattered the Brahmanical monopoly on idol consecration and temple worship, preaching his famous doctrine of “One Caste, One Religion, One God for mankind.”
- **B.R. Ambedkar:** He founded the **Bahishkrit Hitakarini Sabha** in Bombay in 1924. Its central motto was “Educate, Agitate, Organise.” The organisation aimed to promote education, socio-economic upliftment, and political awareness among the Depressed Classes.

Q 36. Ans. B

Explanation:

- The ‘Chowkidari Tax’ boycott was one of the prominent regional manifestations of the Civil Disobedience Movement (1930–34) and was mainly witnessed in Bengal and Bihar. The tax was levied to maintain village watchmen (chowkidars), who were widely perceived as agents of the colonial administration. Villagers refused to pay the tax as a form of non-cooperation with the British government.
- The movement was particularly intense in the Midnapore district of Bengal and several districts of Bihar, where people organised mass campaigns against tax collection.
- **Additional Facts:**
 - ✓ Madras Presidency witnessed important Civil Disobedience activities such as the Vedaranyam Salt March led by C. Rajagopalachari, rather than the Chowkidari Tax boycott.
 - ✓ Bombay Presidency was notable for the Dharasana Salt Satyagraha, forest law violations, and no-tax campaigns in selected areas.
 - ✓ The North-West Frontier Province became famous for the Khudai Khidmatgar (Red Shirt) Movement led by Khan Abdul Ghaffar Khan and the Qissa Khwani Bazaar incident.

Q 37. Ans. B

Explanation:

- **Pair 1 is correctly matched:** The **Dutch East India Company (VOC)** established its first factory in India at **Masulipatnam** (in modern-day Andhra Pradesh)

in **1605**. They were drawn to the Coromandel coast primarily for the lucrative trade in Indian textiles, which they used to barter for spices in the Indonesian archipelago.

- **Pair 2 is correctly matched:** The **English** established their first permanent, legally sanctioned factory at **Surat** in **1613**. Emperor Jahangir issued a farman permitting them to build it after Captain Thomas Best impressed the Mughals by defeating a Portuguese naval fleet at the Battle of Swally (1612).
- **Pair 3 is incorrectly matched:** The **French** did not establish their first factory at Pondicherry. Their first factory was actually set up at **Surat** in **1668** by François Caron. Pondicherry was founded several years later, in 1674, by François Martin, and while it eventually became the nerve centre and headquarters of French power in India, it was not their first settlement.

Q 38. Ans. A

Explanation:

- The Swadesh Bandhab Samiti was founded by Ashwini Kumar Dutt at Barisal (now in Bangladesh) during the Swadeshi Movement following the Partition of Bengal (1905). The organisation played a crucial role in promoting Swadeshi goods, national education, arbitration courts, village upliftment, and the boycott of foreign products. It became one of the most influential voluntary organisations of the movement, especially in rural eastern Bengal.
- **Additional Facts:**
 - ✓ Aurobindo Ghosh emerged as one of the leading ideologues of the extremist movement and was associated with the newspapers *Bande Mataram* and *Karmayogin*, but he did not found the Swadesh Bandhab Samiti.
 - ✓ Rabindranath Tagore promoted the Swadeshi Movement through *Raksha Bandhan* ceremonies, patriotic songs such as ‘Amar Sonar Bangla’, and campaigns for Hindu-Muslim unity.
 - ✓ Surendranath Banerjea was a prominent moderate leader who actively opposed the Partition of Bengal through public meetings, petitions, and political mobilisation, but was not the founder of the Samiti.

Q 39. Ans. A

Explanation:

- **Assertion (A) is true:** In February 1922, Gandhi called off the Non-Cooperation Movement,

suspending the countrywide mass campaign that had been running since 1920, much to the surprise and disappointment of many Congress leaders such as Motilal Nehru and C.R. Das.

- **Reason (R) is true:** On 4–5 February 1922, at Chauri Chaura in the United Provinces (present-day Uttar Pradesh), an angry crowd of protesters clashed with the police and set fire to a police station, killing about 22 policemen trapped inside.
- **Why (R) is the correct explanation of (A):**
- Gandhi held non-violence to be not merely a tactic but a non-negotiable moral foundation of Satyagraha.
- The Chauri Chaura violence convinced him that the movement had strayed from this principle and that the masses were not yet sufficiently disciplined in non-violent methods to be trusted with a mass civil disobedience campaign.
- He therefore withdrew the movement to prevent further violence and to re-emphasise non-violence as central to the freedom struggle, making R the direct cause of A.

Q 40. Ans. B

Explanation:

- The Royal Indian Navy (RIN) Ratings' Mutiny began on 18 February 1946 aboard HMIS Talwar, a shore establishment and signal training school in Bombay (Mumbai). The immediate causes included poor food, racial discrimination, low pay, and ill-treatment by British officers, but the movement soon assumed a broader nationalist character with demands for the release of INA prisoners and the withdrawal of British rule.
- The mutiny rapidly spread from HMIS Talwar to over 70 ships and numerous shore establishments, involving nearly 20,000 naval ratings across Bombay, Karachi, Calcutta, Madras, and other ports.

Q 41. Ans. B

Explanation:

- The All India Harijan Sevak Sangh was founded by Mahatma Gandhi in 1932 following the Poona Pact. The organisation was established to work for the eradication of untouchability and the social, educational, and economic upliftment of the Depressed Classes (later known as the Scheduled Castes). Initially known as the All India Anti-Untouchability League, it was soon renamed the Harijan Sevak Sangh.

● Additional Facts:

- ✓ Dr. B.R. Ambedkar was the foremost leader of the Depressed Classes and founded organisations such as the Bahishkrit Hitakarini Sabha (1924) and the Scheduled Castes Federation (1942), but he did not establish the Harijan Sevak Sangh.
- ✓ Jyotirao Phule founded the Satyashodhak Samaj (1873) to challenge caste discrimination and promote social equality.
- ✓ E.V.R. Periyar founded the Self-Respect Movement (1925) in the Madras Presidency to oppose caste hierarchy and Brahmanical dominance.

Q 42. Ans. B

Explanation:

- **Pair 1 is correctly matched:** Cornwallis's reforms (1793) separated revenue administration from judicial administration by removing judicial powers from the Collector and vesting them in a separate District Judge, ending the earlier practice of the same official collecting revenue and adjudicating disputes.
- **Pair 2 is incorrectly matched:** The maximum age limit for the ICS examination had already been lowered from 21 to 19 in 1878. The Aitchison Commission (1886) actually recommended raising it again, to 23, along with proposals to hold simultaneous examinations in India and create a Statutory Civil Service with greater Indian representation. So the statement reverses the direction of the recommendation.
- **Pair 3 is correctly matched:** The Peel Commission (1859), set up after the Revolt of 1857, reorganised the Indian Army on divide-and-balance lines — increasing the ratio of European to Indian troops and recruiting selectively from communities later labelled "martial races" (Punjabis, Gorkhas, Sikhs), while reducing reliance on regions seen as centres of the 1857 revolt (like Awadh and Bihar).

Q 43. Ans. A

Explanation:

- The Surat Session of the Indian National Congress (1907) was presided over by Rash Behari Ghosh. The session is remembered for the split between the Moderates and the Extremists, leading to the division of the Congress into two factions.
- The Moderates supported Rash Behari Ghosh for the presidency, while the Extremists, led by Bal

Gangadhar Tilak, preferred Lala Lajpat Rai (who later withdrew his candidature). Disagreements over the methods of the national movement, the implementation of the Swadeshi programme, and the choice of the Congress president culminated in a dramatic split at Surat.

Q 44. Ans. D

Explanation:

- **Lucknow Pact (1916) → 3 (Accepted separate electorates for Muslims):** In this Congress-Muslim League agreement, the Congress, led by figures like Tilak, formally accepted the principle of separate electorates for Muslims — a major concession — in exchange for League support on constitutional reforms and self-government.
- **Simon Commission (1927) → 4 (Recommended abolition of dyarchy):** The all-European Simon Commission's 1930 report recommended scrapping the system of dyarchy introduced in the provinces under the Government of India Act, 1919, and replacing it with full responsible government in the provinces, while retaining central control at the centre.
- **Nehru Report (1928) → 1 (Proposed dominion status):** Drafted by a committee under Motilal Nehru in response to the Simon Commission's Indian exclusion, the report's central proposal was that India should be granted Dominion Status on the pattern of Canada or Australia, rather than full independence.
- **Fourteen Points (1929) → 2 (Demanded one-third Muslim representation):** Jinnah's Fourteen Points, framed as a Muslim League response to the Nehru Report, demanded that Muslims be guaranteed no less than one-third representation in the central legislature, among other safeguards.

Q 45. Ans. B

Explanation:

- The Eka (Unity) Movement (1921–22) was a peasant movement that primarily emerged in the Awadh region of the United Provinces (present-day Uttar Pradesh). It developed after the decline of the Awadh Kisan Sabha and sought to unite peasants against high rents, illegal cesses, forced labour (begar), and oppressive landlord practices. The Eka Movement followed the Awadh Kisan Sabha Movement led by Baba Ram Chandra. It was concentrated mainly in the districts of Hardoi, Bahraich, Sitapur, and Barabanki.

- The movement was led by local leaders such as Madari Pasi and attracted tenants, small peasants, and agricultural labourers. Participants took a solemn oath (eka) to refuse excessive rents, pay only the recorded rent, and remain united against exploitation. Although it initially had support from Congress and Khilafat workers, it later became more militant and eventually lost official nationalist backing.

Additional Facts:

- ✓ The Malabar Coast witnessed the Moplah (Mappila) Rebellion (1921).
- ✓ The Bengal Presidency was associated with movements such as the Swadeshi Movement and later the Tebhaga Movement.
- ✓ The Andhra Delta witnessed peasant agitations and the Ryot movements, but not the Eka Movement.

Q 46. Ans. D

Explanation:

- The Indian Councils Act of 1892 was the first legislation to grant members of the Legislative Councils the right to discuss the annual financial budget. Although members could debate the budget, they could neither vote on it nor move resolutions or demand grants, making the reform largely advisory in nature.
- The Act also modestly expanded the Legislative Councils and introduced the principle of indirect election through recommendations by provincial bodies such as municipalities, district boards, universities, and chambers of commerce, though the term "election" was deliberately avoided in the Act.
- **Additional Facts:**
 - ✓ The Indian Councils Act of 1861 restored legislative powers to the Bombay and Madras Presidencies and introduced the system of nominating Indians to Legislative Councils, but it did not permit budget discussions.
 - ✓ The Indian Councils Act of 1909 (Morley-Minto Reforms) further enlarged the councils, introduced separate electorates for Muslims, and allowed members to move resolutions and ask supplementary questions.
 - ✓ The Government of India Act of 1919 (Montagu-Chelmsford Reforms) introduced dyarchy in the provinces, substantially enlarged legislative powers, and enabled voting on parts of the budget.

Q 47. Ans. A

Explanation:

- **Vernacular Press Act (1878):** Enacted by Lord Lytton, the Act sought to curb the freedom of the Indian-language press by empowering the government to confiscate printing presses publishing material deemed seditious. It was repealed by Lord Ripon in 1882.
- **First Factory Act (1881):** Passed during the tenure of Lord Ripon, it was the first Indian factory legislation, enacted mainly to regulate the working conditions of children in factories. It prohibited the employment of children below 7 years, limited the working hours of children, and provided for factory inspections.
- **Government Resolution on Local Self-Government (1882):** Issued by Lord Ripon, this landmark resolution laid the foundation of local self-government in India by promoting elected local bodies and greater participation of Indians in municipal and district administration. Lord Ripon is therefore regarded as the “Father of Local Self-Government in India.”
- **Establishment of the Indian National Congress (1885):** The Indian National Congress (INC) was founded in 1885 under the initiative of A.O. Hume. Its first session was held at Bombay under the presidency of W.C. Bonnerjee, with 72 delegates participating.

Q 48. Ans. A

Explanation:

- Bhulabhai Desai played the leading role in defending the officers of the Indian National Army (INA) during the famous Red Fort Trials (1945–46). As the chief counsel of the defence team, he argued that the INA soldiers were patriots fighting for India’s independence rather than traitors guilty of waging war against the British Crown.
- The first trial involved Shah Nawaz Khan, Prem Kumar Sehgal, and Gurbaksh Singh Dhillon, representing India’s three major religious communities. The defence team also included prominent lawyers such as Jawaharlal Nehru, K.N. Katju, Asaf Ali, and Tej Bahadur Sapru, but Bhulabhai Desai was the principal architect of the legal defence.

Q 49. Ans. A

Explanation:

- **Vellore Mutiny (1806):** The Vellore Mutiny was the first major military uprising against the British East India Company. It broke out on 10 July 1806 at the

Vellore Fort due to resentment over new military dress regulations and fears of religious interference. Although swiftly suppressed, it foreshadowed the Revolt of 1857.

- **Treaty of Amritsar (1809):** Signed between the British East India Company and Maharaja Ranjit Singh, the treaty recognised Ranjit Singh’s sovereignty west of the Sutlej River, while he agreed not to expand his authority east of the Sutlej. It established friendly relations between the Sikh Empire and the British.
- **Third Anglo-Maratha War (1817–1818):** This war resulted in the final defeat of the Maratha Confederacy. The office of the Peshwa was abolished, Baji Rao II was pensioned off to Bithoor, and the British emerged as the paramount power in India.
- **First Burmese War (1824–1826):** Fought between the British East India Company and the Konbaung Dynasty of Burma, the war ended with the Treaty of Yandabo (1826). Burma ceded Assam, Manipur, Arakan, and Tenasserim to the British and agreed to pay a heavy indemnity.

Q 50. Ans. C

Explanation:

- **Appointment of the Harcourt Butler Committee (1927):** The Indian States Committee, popularly known as the Harcourt Butler Committee, was appointed by the British Government to examine the relationship between the Indian States (Princely States) and the British Crown. It is recommended that the relationship between the Crown and the princely states should not be transferred to any future responsible government in British India without the consent of the rulers.
- **Lahore Session of the Indian National Congress (1929):** Held under the presidency of Jawaharlal Nehru, the session adopted the historic Purna Swaraj (Complete Independence) resolution. It also declared that 26 January 1930 would be observed as Independence Day throughout the country.
- **Dandi March (1930):** On 12 March 1930, Mahatma Gandhi commenced the Salt March from Sabarmati Ashram to Dandi, covering about 390 km (240 miles) in 24 days. By breaking the Salt Law on 6 April 1930, Gandhi launched the Civil Disobedience Movement.
- **Gandhi-Irwin Pact (1931):** Signed on 5 March 1931 between Mahatma Gandhi and Lord Irwin, the pact led to the suspension of the Civil Disobedience Movement. The government agreed to release political prisoners (except those convicted of

violence) and permit peaceful manufacture of salt, while Gandhi agreed to participate in the Second Round Table Conference.

Q 51. Ans. D

Explanation:

- The Statutory Liquidity Ratio (SLR) is a prudential measure under which (as per the Banking
- Under the Regulations Act 1949, all Scheduled Commercial Banks in India must maintain an amount in one of the following forms as a percentage of their total Demand and Time Liabilities (DTL) / Net DTL (NDTL):
 - ✓ **Cash Gold, or Investments in unencumbered (i.e. debt-free) Instruments that include:**
 - ✓ Treasury Bills of the Government of India
 - ✓ Dated securities, including those issued by the Government of India from time to time under the market borrowings program and the Market Stabilisation Scheme (MSS)
 - ✓ State Development Loans (SDLs) issued by State Governments under their market borrowings program
 - ✓ Other instruments as notified by the RBI.
- **Corporate bonds** issued by private companies are not eligible SLR assets.

Q 52. Ans. B

Explanation:

- The Eight Core Industries (ECI) are identified by the Ministry of Commerce and Industry and together account for about 40.27% of the weight of the Index of Industrial Production (IIP). These industries are considered the backbone of the industrial economy.
- **The Eight Core Industries are:**
 - ✓ Coal
 - ✓ Crude Oil
 - ✓ Natural Gas
 - ✓ Refinery Products
 - ✓ Fertilisers
 - ✓ Steel
 - ✓ Cement
 - ✓ Electricity
- Automobiles are part of the manufacturing sector but are not classified as a core industry.
- Real Estate is a service/economic activity and is not included among the Eight Core Industries.

Q 53. Ans. B

Explanation:

- **Statement 1 is correct:** The inflationary gap is

a macroeconomic concept that measures the difference between the current level of real Gross Domestic Product (GDP) and the potential GDP if the economy were operating at full capacity (full employment level). When the current real GDP exceeds this potential GDP, it indicates that the economy is overheating.

- **Statement 2 is incorrect:** An inflationary gap is caused by an **excess** of aggregate demand, not a deficiency. When the total demand in an economy surpasses the total supply capacity at the full employment level, producers cannot increase output further to meet it. This excess demand puts upward pressure on prices, leading to demand-pull inflation.

Q 54. Ans. C

Explanation:

- Reflation refers to a set of expansionary monetary and fiscal policy measures adopted by the government or the central bank to stimulate economic activity, increase aggregate demand, and move the economy out of a recession or period of low inflation.
 - ✓ **Statement 1 is correct:** Increase in public expenditure boosts aggregate demand, employment, and investment, making it a key reflationary measure.
 - ✓ **Statement 2 is correct:** Reduction in interest rates by the central bank encourages borrowing and investment while stimulating consumption, thereby supporting reflation.
 - ✓ **Statement 3 is incorrect:** Increase in tax rates is a contractionary fiscal measure. It reduces disposable income and aggregate demand, working against the objective of reflation.
 - ✓ **Statement 4 is correct:** Expansionary fiscal policy, involving higher government spending and/or lower taxes, is one of the principal instruments used to achieve reflation.

Q 55. Ans. B

Explanation:

- **Statement 1 is correct:** Non-Banking Financial Companies (NBFCs) are companies registered under the Companies Act (earlier the Companies Act, 1956, and now governed by the Companies Act, 2013) and regulated by the Reserve Bank of India (RBI) under the Reserve Bank of India Act, 1934. They are engaged in financial activities such as loans and advances, acquisition of shares, leasing, hire-purchase, insurance business, and investment activities.

- **Statement 2 is incorrect:** NBFCs cannot issue cheques drawn on themselves, as they do not possess a banking licence. Unlike banks, NBFCs cannot accept demand deposits, are not part of the payment and settlement system, and therefore cannot provide cheque-book facilities to customers.

- **Additional Facts:**

- ✓ NBFCs cannot accept demand deposits, whereas banks can.
- ✓ NBFCs cannot issue cheques or participate in the payment and settlement system.
- ✓ Unlike banks, deposits with NBFCs are not covered under the Deposit Insurance and Credit Guarantee Corporation (DICGC).
- ✓ Although NBFCs perform many bank-like financial functions, they are not classified as banks and are regulated separately by the RBI.

Q 56. Ans. C

Explanation:

- The Capital Adequacy Ratio (CAR), also known as the Capital to Risk-Weighted Assets Ratio (CRAR), measures a bank's ability to absorb losses and protect depositors. It indicates whether a bank has sufficient capital in relation to the risks associated with its assets.
- **The formula is:**
 - ✓ $CAR = \frac{\text{Tier I Capital} + \text{Tier II Capital}}{\text{Risk-Weighted Assets (RWAs)}} \times 100$
 - ✓ **Tier I Capital (Core Capital):** Includes equity capital, disclosed reserves, and retained earnings. It is the primary source for absorbing losses while the bank continues its operations.
 - ✓ **Tier II Capital (Supplementary Capital):** Includes subordinated debt, revaluation reserves, general provisions, and hybrid instruments. It provides additional protection in case of liquidation.
 - ✓ **Risk-Weighted Assets (RWAs):** Assets are assigned different risk weights based on their credit risk. For example, government securities generally carry a lower risk weight than commercial loans.

Q 57. Ans. D

Explanation:

- **Hard currency:** It refers to stable, highly liquid, and globally traded international currencies (like the US Dollar or Euro) in which high faith is shown and which are in demand worldwide.
- **Heated currency:** This denotes a domestic currency

experiencing significant depreciation pressure, often caused by a rapid outflow of hard or "hot" money from the economy.

- **Dear money:** Also known as "tight money," this term refers to a financial environment or monetary policy where borrowing becomes expensive due to relatively higher interest rates

Q 58. Ans. A

Explanation:

- Pigovian Tax is a tax imposed on goods or activities that generate negative externalities (costs imposed on society that are not borne by the producer or consumer). The objective is to internalise the external cost by making the polluter or user bear the social cost of the activity.
- Tax on Diesel
 - ✓ Diesel consumption contributes to air pollution, greenhouse gas emissions, and adverse public health effects. Imposing a higher tax on diesel discourages excessive consumption and helps account for these external costs. Hence, it is the best example of a Pigovian Tax.

Q 59. Ans. C

Explanation:

- According to the Economic Survey 2025–26, citing a comprehensive joint assessment by DPIIT and NCAER, India's logistics cost has officially dropped to 7.97% of GDP. This decline is attributed to the tangible payoffs from integrated, cross-sector infrastructure initiatives such as PM GatiShakti, the Dedicated Freight Corridors, Bharatmala, and the broader National Logistics Policy.

Q 60. Ans. C

Explanation:

- The Effective Revenue Deficit (ERD) is a refinement of the Revenue Deficit concept. It recognises that a part of the revenue expenditure consists of grants given to States or other implementing agencies for the creation of capital assets. Since these grants ultimately result in asset creation, they are excluded while calculating the Effective Revenue Deficit.
- **The formula is:**
 - ✓ $\text{Effective Revenue Deficit} = \text{Revenue Deficit} - \text{Grants for Creation of Capital Assets}$
- **Where:**
 - ✓ $\text{Revenue Deficit} = \text{Revenue Expenditure} - \text{Revenue Receipts}$
 - ✓ Grants for Creation of Capital Assets include

transfers made by the Central Government to States, Union Territories, or other agencies for building roads, schools, hospitals, irrigation projects, and other capital assets.

Q 61. Ans. C

Explanation:

- **Yıldırımhan missile:** The Yıldırımhan is Türkiye's first domestically developed intercontinental ballistic missile (ICBM), unveiled in May 2026. It is designed by the research and development center of the Turkish Ministry of National Defence (MSB ARGE) and represents a major step in Turkey's strategic autonomy.
- **DAMPE satellite:** The Dark Matter Particle Explorer (DAMPE), also nicknamed "Wukong," is China's first astronomical satellite. Launched in 2015, it is a space-based telescope designed to detect high-energy gamma rays, electrons, and cosmic rays in the search for evidence of dark matter.
- **Oreshnik hypersonic missile:** The Oreshnik is an operational Russian intermediate-range ballistic missile (IRBM) that gained international attention in late 2024. It is characterised by high speeds (exceeding Mach 10) and the capability to carry multiple independently targetable reentry vehicles (MIRV).

Q 62. Ans. C

Explanation:

- The first satellite-tagged Ganges softshell turtle (*Nilssonina gangetica*) was released into the waters of Kaziranga National Park and Tiger Reserve in Assam. The initiative was a collaborative effort involving the Wildlife Institute of India (WII), the Assam Forest Department, and the Turtle Survival Alliance, aimed at studying the species' movement patterns and habitat requirements.
- **Additional Facts:**
 - ✓ The release took place in May 2026 to coincide with Endangered Species Day.
 - ✓ The Ganges softshell turtle is classified as 'Endangered' by the IUCN and is legally protected under Schedule I of the Wildlife (Protection) Act, 1972.
 - ✓ Kaziranga was chosen for this conservation milestone due to its extensive riverine ecosystems, which provide a critical habitat for this species.

Q 63. Ans. D

Explanation:

- The Indian Army has declared 2026 as the "Year of Networking and Data Centricity."
- This initiative, announced by the previous Chief of the Army Staff, General Upendra Dwivedi, aims to accelerate the force's digital transformation. By prioritising integrated communication networks and real-time data-driven decision-making, the Army seeks to enhance its combat effectiveness, cyber resilience, and agility for modern, technology-driven warfare.
- This designation follows the 2024–25 "Year of Technology Absorption," representing the next phase in the Army's ongoing decade of transformation.

Q 64. Ans. A

Explanation:

- iLive Connect is India's first doctor-led, AI-enabled healthcare ecosystem designed for continuous, round-the-clock remote clinical monitoring. It bridges the gap between hospital-based care and home recovery by using medical-grade wearable biosensors (such as a chest patch and wristband) that transmit real-time vital data, like ECG, heart rate, oxygen levels, and blood pressure, to a central medical command centre staffed by specialists.

Q 65. Ans. A

Explanation:

- **Mobile World Congress 2026:** Held in Barcelona, Spain, its theme was "The IQ Era." This focus emphasised the convergence of artificial intelligence with global connectivity, exploring how AI-driven networks and "intelligent infrastructure" are transforming business, industry, and daily life.
- **Biofach 2026:** Hosted in Nuremberg, Germany, the world's leading organic trade fair operated under the theme "Growing Tomorrow: Young Voices, Bold Visions." This theme highlighted the role of the next generation of changemakers in reimagining sustainable food systems and organic farming practices within planetary boundaries.
- **Munich Security Conference 2026:** Centred in Munich, Germany, the 62nd edition of this premier security summit was guided by the theme "Under Destruction." This title served as the overarching framework for the Munich Security Report 2026, which analysed the active erosion of the post-1945

international order and the challenges posed by a fragmenting, multipolar world.

- **Global AI Show 2025:** Taking place in Abu Dhabi, UAE, this event focused on the theme “AI-2031: Accelerating Intelligent Futures.” It brought together global leaders, innovators, and investors to discuss the next decade of AI development, focusing on governance frameworks, sustainable AI integration, and the practical application of generative AI across vital sectors.

Q 66. Ans. D

Explanation:

- **Pair 1 is correctly matched:** Goa became the second state in India, after Kerala, to release a comprehensive state-level Bird Atlas to document avian biodiversity and support ecological monitoring.
- **Pair 2 is correctly matched:** Andhra Pradesh is set to become the first state to receive the ‘PRAMAAN’ certification, an accreditation by the Central Government under the Indian Forest and Wood Certification Scheme (IFWCS), recognising environmentally sustainable cultivation practices for forest products.
- **Pair 3 is correctly matched:** Kerala became the first state in India to pass a bill amending the Central Wildlife Protection Act, 1972, with the objective of addressing and reducing growing human-animal conflicts.
- **Pair 4 is correctly matched:** Chandigarh achieved a 99.93% literacy rate under the New India Literacy Programme, making it the second fully literate Union Territory in India after Ladakh.

Q 67. Ans. D

Explanation:

- **Statement 1 is correct:** India (ISRO) signed an agreement with the Swedish Institute of Space Physics for the collaborative study of Venus under the Shukrayaan-1 mission.
- **Statement 2 is correct:** Bharat Electronics Limited (BEL) and Safran (France) partnered to manufacture the HAMMER missile in India under the ‘Make in India’ initiative. The weapon system was originally developed by Safran.
- **Statement 3 is correct:** In February 2026, UIDAI and Google joined hands to enable the display of authorised Aadhaar centres on Google Maps to enhance access and convenience for residents across the country.

- **Statement 4 is correct:** The Punjab Energy Development Agency (PEDA) signed a Memorandum of Understanding with the Indian Institute of Science (IISc), Bengaluru, to establish a pilot demonstration project producing green hydrogen from agricultural biomass, specifically paddy (rice) straw.

Q 68. Ans. B

Explanation:

- **She-Box Portal** (Sexual Harassment Electronic Box) is a centralised digital governance platform launched by the Ministry of Women and Child Development. It is designed to streamline the complaint registration process for workplace-related sexual harassment cases under the PoSH Act.
- **e-SEHAT App** was launched by the Chief Minister of Jammu and Kashmir to cater to the healthcare needs of the Union Territory’s citizens. It is a digital application designed to deliver accessible healthcare facilities, including teleconsultations and emergency care, to patients in both the private and public sectors.
- **e-Shram Saathi App** was launched by the Government of Chhattisgarh as a ‘digital labour chowk’. It connects workers with employment opportunities and government welfare schemes through a single platform that they can access from their homes.
- **PRASHAST 2.0** (Pre-Assessment Holistic Screening Tool) is an enhanced digital platform by the National Council of Educational Research and Training (NCERT). It facilitates the early detection of 21 types of disabilities in school children to ensure they receive timely assessment and support.

Q 69. Ans. C

Explanation:

- In April 2026, Mumbai Falcons Racing Limited (MFRL) appointed Indian cricketer **Shreyas Iyer** as the official brand ambassador for the Formula 1 Program in India. He was selected to represent the launch of the F1 Sim Racing India Open 2026, an official Formula 1-sanctioned nationwide competition aimed at creating a structured platform for emerging esports and real-world motorsport talent across the country.
- **Mahendra Singh Dhoni:** In March 2026, MS Dhoni was announced as the brand ambassador for the BHIM Payments App by the NPCI to promote

digital payment accessibility and trust across Indian households. Additionally, in June 2026, he was appointed as the National Brand Ambassador for DRiV, a major global automotive aftermarket parts manufacturer.

- **Sanju Samson:** He serves as the official ambassador for the English Premier League (EPL) in India, marking a unique cross-sport connection to help promote football's growth and fan engagement in the country. He is also the brand ambassador for the Kerala-based Adi Group of Institutions.
- **Rohit Sharma:** In November 2025, former Indian T20 captain Rohit Sharma was officially named the tournament ambassador for the ICC Men's T20 World Cup 2026, which will be co-hosted by India and Sri Lanka.

Q 70. Ans. B

Explanation:

- In September 2025, Marium Fatima, a young chess prodigy from Muzaffarpur, created history by becoming the first female player from Bihar to be awarded the Woman FIDE Master (WFM) title by the International Chess Federation (FIDE).
- She achieved this significant milestone after a brilliant performance at the Barbera del Valles Open Chess Championship in Spain, securing the 2100 rating points required to earn this prestigious women-only international chess designation

Q 71. Ans. B

Explanation:

- **Statement 1 is incorrect:** Anju Rathi Rana made legal history in India by becoming the first woman to be appointed as the Law Secretary in the Department of Legal Affairs, under the Ministry of Law and Justice.
- **Statement 2 is correct:** NASA appointed David Salvagnini as its first Chief Artificial Intelligence Officer. In this role, he is tasked with aligning the strategic vision for AI and guiding the agency's responsible use of AI tools.
- **Statement 3 is incorrect:** Nivedita (Nibedita) Dubey became the first woman officer to be appointed as a Member (Human Resources) on the Airports Authority of India (AAI) Board.
- **Statement 4 is correct:** Suman Kumari passed her judicial exams in 2019 to become the first Hindu woman to be appointed as a civil judge in Pakistan.

Q 72. Ans. B

Explanation:

- **Pair 1 is correctly matched:** The Uttarakhand government launched "Operation Kalanemi". This operation aims to crack down on fake sadhus who deceive the public and exploit religious identities.
- **Pair 2 is correctly matched:** The Kerala Police initiated "Operation Toofan". This anti-drug project is aimed at eradicating narcotics networks and finding the source of drugs coming into the state.
- **Pair 3 is incorrectly matched:** "Operation Surakshit Cyberspace" was launched by the Gujarat Police, not Bihar. It is a 28-day statewide campaign designed to protect women and children online.
- **Pair 4 is incorrectly matched:** "Operation Gang Bust" was led primarily by the Delhi Police, not Punjab. It involved a major 48-hour crackdown operation across multiple Indian state police agencies.

Q 73. Ans. B

Explanation:

- In March 2026, the Assembly of Experts elected Mojtaba Khamenei as the third Supreme Leader of Iran following the assassination of his father, Ayatollah Ali Khamenei, during the 2026 Iran war.
- His appointment was based on a decisive vote by the 88-member clerical body. Khamenei's succession highlighted his deep historical ties to the paramilitary Islamic Revolutionary Guard Corps (IRGC) and indicated that hardline factions retain strong control within the Iranian establishment.

Q 74. Ans. B

Explanation:

- The 16th edition of the India-US Joint Special Forces Exercise "Vajra Prahar" was conducted at the Special Forces Training School in **Bakloh, Himachal Pradesh**, from February 24 to March 16, 2026.
- The primary aim of the exercise is to promote robust military cooperation between the Indian Army's Special Forces and the US Army's Green Berets. The training focuses on enhancing interoperability, joint mission planning, and the mutual exchange of tactical expertise for special operations in mountainous terrain.

Q 75. Ans. A

Explanation:

- **Pair 1 is correctly matched:** In August 2025, Moldova officially joined the International Solar

Alliance (ISA) as its 107th member, committing to sustainable solar energy projects and climate change mitigation.

- **Pair 2 is incorrectly matched:** In June 2026, **Botswana**, not Ghana, became the 68th nation to sign the Artemis Accords, an international agreement led by NASA that outlines principles for the peaceful and sustainable exploration of space.
- **Pair 3 is incorrectly matched:** In May 2026, **Saudi Arabia**, not the UAE, officially joined the India-led International Big Cat Alliance (IBCA) as its 26th member nation to support global efforts in conserving the world's principal big cats.

Q 76. Ans. B

Explanation:

- **Pair 1 is correctly matched:** Namakkal Soft Stone Cookware (locally called kalchatti or makkal pathirangal), a soapstone cookware craft from Namakkal district, received its GI tag, becoming one of five new Tamil Nadu products (along with Woraiyur cotton sari, Kavindapadi jaggery, Thooyamalli rice, and Ambasamudram wooden toys) to take the state's GI tally to 74.
- **Pair 2 is correctly matched:** Rignai Pachara, a traditional hand-woven attire worn by indigenous tribal women of Tripura, received the GI tag along with Matabari Pera Prasad (the milk-sugar sweet offering at the Tripureswari temple).
- **Pair 3 is incorrectly matched:** Ajrakh block-printing does have a presence in Barmer, Rajasthan, but the GI-tagged product is specifically "Kutch Ajrakh," registered to Gujarat, granted to artisans of the Ajrakhpur and Dhamadka villages in Kutch district. Rajasthan's Barmer-based Ajrakh printing has not received a separate GI certification.

Q 77. Ans. A

Explanation:

- **Statement 1 is correct:** The 2025 G20 Summit (the 20th G20 leaders' meeting) was held on 22–23 November 2025 at the Johannesburg Expo Centre, South Africa, the first-ever G20 summit hosted on African soil.
- **Statement 2 is correct:** South Africa's presidency ran on the theme "Solidarity, Equality, Sustainability," built around fairness in global development and long-term environmental responsibility. This matches the substance of "Building a Just World and a Sustainable Planet"
- **Statement 3 is incorrect:** The African Union did not

become a permanent G20 member for the first time in 2025. It was admitted as a permanent member back in 2023, at the New Delhi Summit under India's G20 presidency. The 2025 Johannesburg summit was instead notable as the first G20 summit held since the AU joined, not the occasion of its induction.

Q 78. Ans. B

Explanation:

- **Pair 1 is incorrectly matched:** Exercise Mitra Shakti is a joint military exercise between India and Sri Lanka, not the Maldives, first held in 2012, and conducted alternately in both countries. Its 11th edition (2025) was held at Belagavi, Karnataka, and focused on counter-terrorism and sub-conventional operations under UN Chapter VII.
- **Pair 2 is correctly matched:** Exercise Harimau Shakti is a bilateral Army exercise between India and Malaysia. Its 5th edition was held at the Mahajan Field Firing Range, Rajasthan, involving India's Dogra Regiment and Malaysia's 25th Battalion, Royal Malaysian Army.
- **Pair 3 is correctly matched:** Exercise Samudra Shakti is a bilateral naval exercise between India and Indonesia, initiated in 2018 under India's Act East Policy. Its 5th edition was hosted by the Indian Navy at Visakhapatnam.

Q 79. Ans. C

Explanation:

- **Pair 1 is correctly matched:** "The Library Man of India: The Story of P.N. Panicker" is authored by P.P. Sathyan (with inputs from N. Balagopal and Rajeev Nair), published by Rupa Publications. Vice President C.P. Radhakrishnan only released the book at Uprashtrapati Bhavan
- **Pair 2 is correctly matched:** "Better Never Stops" is authored by Shyam Srinivasan, former MD & CEO of Federal Bank, a part-memoir and part-leadership guide drawing on his three-decade banking career.
- **Pair 3 is correctly matched:** "Karuna: The Power of Compassion" is authored by Kailash Satyarthi, the 2014 Nobel Peace Laureate, published by HarperCollins India and launched at the Jaipur Literature Festival in January 2026.
- **Pair 4 is incorrectly matched:** "Tides of Time: Bharat's History through Murals in Parliament" is authored by Sudha Murty (Rajya Sabha MP), a Lok Sabha Secretariat publication interpreting 124 murals in Parliament.

Q 80. Ans. C

Explanation:

- **Statement 1 is correct:** At the seventh edition of the Sportstar Aces Awards (Taj Mahal Palace, Mumbai), Manu Bhaker, a double medallist at the Paris 2024 Olympics, was named both “Sportstar of the Year (Female)” and “Sportswoman of the Year in an Olympic Sport (Female).”
- **Statement 2 is correct:** Ousmane Dembélé of Paris Saint-Germain and France won the 2025 Men’s Ballon d’Or, awarded at Théâtre du Châtelet, Paris, ahead of runner-up Lamine Yamal. His award followed a stellar 2024–25 season in which he helped PSG win their first UEFA Champions League title, was named Champions League Player of the Season, and finished with 33 goals and 15 assists across competitions.

Q 81. Ans. D

Explanation:

- **Statement 1 is correct:** According to RBI’s weekly statistical supplement, India’s forex reserves rose to \$703.31 billion for the week ended 17 April 2026, up \$2.36 billion from the previous week following a dip from the all-time high of \$728.49 billion touched in February 2026.
- **Statement 2 is correct:** The Economic Survey 2025–26, tabled by Finance Minister Nirmala Sitharaman, confirmed that India remained the world’s largest recipient of remittances, with inflows reaching \$135.4 billion in FY25, a trend it has maintained since 2008, per World Bank data.
- **Statement 3 is correct:** As per the National Statistical Office’s official release, India’s retail (CPI) inflation stood at 3.40% for March 2026 (year-on-year), up from 3.21% in February, with rural inflation at 3.63% and urban at 3.11%.

Q 82. Ans. C

Explanation:

- **Jnanpith Award 2025 →2 (R. Vairamuthu):** The 60th Jnanpith Award, India’s highest literary honour, was conferred on Tamil poet, lyricist and writer R. Vairamuthu, announced by the Bharatiya Jnanpith, making him only the third Tamil writer to win it, after Akilan (1975) and D. Jayakanthan (2002).
- **Vyas Samman 2024 →4 (Suryabala):** The 34th Vyas Samman, instituted by the K.K. Birla Foundation, went to Hindi writer Suryabala for her novel Kaun

Des Ko Vasi: Venu Ki Diary, which explores the cultural and identity struggles of Indian youth in America.

- **Saraswati Samman 2025 →3 (Ramkumar Mukhopadhyay):** The 35th Saraswati Samman was awarded to Bengali author Ramkumar Mukhopadhyay for his novel Hara Parbati Katha, ending a 22-year gap since a Bengali writer last won the honour.
- **Dadasaheb Phalke Award 2023 →1 (Mohanlal):** The 55th Dadasaheb Phalke Award, India’s highest cinema honour, was conferred on Malayalam superstar Mohanlal at the 71st National Film Awards ceremony, making him only the second Malayalam film personality to receive it, after Adoor Gopalakrishnan (2004).

Q 83. Ans. D

Explanation:

- **NISAR →1 (Joint ISRO-NASA earth observation satellite):** NISAR, a \$1.5 billion Earth-observation mission jointly developed by NASA and ISRO carrying dual-frequency (L-band and S-band) synthetic aperture radar, was launched aboard ISRO’s GSLV-F16 on 30 July 2025 from Satish Dhawan Space Centre, Sriharikota.
- **Blue Bird Block-2 →2 (Communication satellite launched via LVM3):** ISRO’s LVM3-M6 rocket launched the 6,100-kg BlueBird Block-2 satellite of US-based AST SpaceMobile the heaviest payload ever carried by LVM3, meant to enable direct-to-mobile satellite connectivity.
- **EOS-NI (Anvesha) →3 (Launched under PSLV-C62, unsuccessful):** The DRDO-developed EOS-N1 (Anvesha) hyperspectral Earth-observation satellite was launched aboard PSLV-C62, but the mission failed after a roll-rate anomaly during the third stage ISRO’s second consecutive PSLV failure after PSLV-C61.
- **Vyommitra →4 (AI-powered half-humanoid robot for Gaganyaan):** Vyommitra, developed by ISRO, is an AI-powered half-humanoid robot (head, torso, and arms, no legs) designed to fly aboard the uncrewed Gaganyaan G-1 mission to test life-support systems and crew interfaces ahead of India’s first human spaceflight.

Q 84. Ans. D

Explanation:

- **Statement 1 is correct:** Haven-1, developed by the American company Vast Space, is being built

as the world's first private/commercial space station, targeted to launch on a SpaceX Falcon 9 (the timeline has slipped from earlier 2026 targets toward Q1 2027, but Vast remains the developer building toward this milestone).

- **Statement 2 is correct:** Bengaluru-based startup GalaxEye successfully launched Mission Drishti aboard a SpaceX Falcon 9 — the world's first OptoSAR satellite, combining Electro-Optical (EO) and Synthetic Aperture Radar (SAR) sensors on a single operational platform for all-weather Earth imaging.
- **Statement 3 is correct:** Agnikul Cosmos successfully test-fired its single-piece, 3D-printed booster engine "Agnite" at its Chennai facility, the world's largest single-piece 3D-printed rocket engine, part of the propulsion system for its Agnibaan launch vehicle.

Q 85. Ans. A

Explanation:

- **Shirley Botchway:** Shirley Ayorkor Botchwey is the 7th Secretary-General of the Commonwealth of Nations, a role she assumed on April 1, 2025. She is a Ghanaian diplomat and politician.
- **Christina Koch** (NASA astronaut), on the Artemis II mission (launched 1 April 2026, lunar flyby 6 April 2026), became the first woman to travel around the Moon and the first woman to travel beyond low Earth orbit.
- **Jennifer Simons:** First Woman President of Suriname (Correct): Jennifer Geerlings-Simons was elected and inaugurated as the President of Suriname on July 16, 2025, becoming the first woman to hold this office in the nation's history.

Q 86. Ans. A

Explanation:

- Bhutan became the first country in the world to launch a national-level cryptocurrency payment system for tourism, in a partnership between the Department of Tourism, Binance Pay, and DK Bank (Bhutan's first fully digital bank). Under this system, international tourists can pay for nearly every aspect of their trip: airline tickets, tourist visas, the Sustainable Development Fee, hotel bookings, tour guides, monument entry fees, and even local shopping — using over 100 cryptocurrencies (including Bitcoin, BNB, and USDC), settled via QR codes and instantly converted into Bhutanese Ngultrum for merchants.

Q 87. Ans. C

Explanation:

- **Pair 1 is correctly matched:** UNESCO (United Nations Educational, Scientific and Cultural Organisation) is headquartered in Paris, France.
- **Pair 2 is incorrectly matched:** UNICEF's (United Nations Children's Fund) headquarters is in New York City, not Geneva. Geneva does host several other UN bodies (such as WHO, ILO, UNHCR, and the UN Office at Geneva), which is likely the source of confusion, but UNICEF itself is New York-based.
- **Pair 3 is correctly matched:** The OECD (Organisation for Economic Co-operation and Development) is headquartered in Paris, France (at the Château de la Muette).
- **Pair 4 is correctly matched:** NATO (North Atlantic Treaty Organisation) is headquartered in Brussels, Belgium.

Q 88. Ans. C

Explanation:

- State Bank of India (SBI) was named the "World's Best Consumer Bank 2025" by Global Finance magazine (New York), along with a second honour, "Best Bank in India 2025." Both awards were presented in Washington, D.C., during the World Bank/IMF Annual Meetings, at the Global Finance World's Best Banks Awards ceremony (now in its 32nd edition, evaluating banks across 150+ countries).
- **SBI was recognised for:**
 - ✓ Its digital-first, consumer-first approach, anchored by its YONO app (100+ million users, ~10 million daily active users)
 - ✓ Leadership in financial inclusion, serving over 520 million customers and adding roughly 65,000 new customers daily

Q 89. Ans. C

Explanation:

- The name of India's first fully indigenous universal multi-calibre rocket launcher system developed for the Indian Army is **Suryastra**.
- The **Suryastra** is a high-precision, long-range rocket artillery system manufactured by Pune-based Nibe Limited in collaboration with Israel's Elbit Systems. Key features of the system include:
 - ✓ **Universal Multi-Calibre Capability:** Designed to fire a wide variety of munitions, including guided rockets and missiles, from a single modular platform.

- ✓ **Precision Strike:** It is capable of executing surface-to-surface precision strikes at ranges extending up to **300 km**.
- ✓ **Mobility and Flexibility:** The launcher is designed to be highly mobile, suitable for deployment in diverse terrains, and capable of “shoot and scoot” tactics to evade counter-battery fire.
- ✓ **Strategic Significance:** It marks a major milestone in India’s “Aatmanirbhar Bharat” (Self-Reliant India) initiative, bridging the gap between conventional artillery and heavy ballistic missiles.

Q 90. Ans. C

Explanation:

- The 2026 Asian Games, officially the 20th Asian Games, will be held in the **Aichi Prefecture and Nagoya City**, Japan.
- **Dates:** September 19 – October 4, 2026.
- **Significance:** This will be the third time Japan hosts the Asian Games (after Tokyo in 1958 and Hiroshima in 1994).
- **Slogan:** “Imagine One Asia.”
- **Key Concept:** The organisers are focusing on an “Athlete-Centred” approach and utilising existing sports facilities across Aichi and Nagoya to keep the games sustainable.
- **Mascot:** The mascot for the Games is named HONOHON.

Edition	Year	Host City	Host Country
19th	2022 (Held 2023)	Hangzhou	China
20th	2026	Aichi-Nagoya	Japan
21st	2030	Doha	Qatar
22nd	2034	Riyadh	Saudi Arabia

Q 91. Ans. B

Explanation:

- **Statement 1 is incorrect:** **plasma** is a state of matter consisting of **ionised gases** containing free electrons and ions. However, plasma is formed at **very high temperatures** or under conditions where a gas becomes ionised by gaining sufficient energy. Examples such as the **Sun, stars, lightning, and fluorescent tubes**, where gases exist in the plasma state due to ionisation. Therefore, the statement is incorrect because it refers to a **very low temperature**, whereas plasma is generally associated with **very**

high temperatures.

- **Statement 2 is correct:** the **Sun and stars** are composed largely of **plasma**, an ionised gaseous state of matter. The extremely high temperatures present in these celestial bodies ionise the gases, producing charged particles that emit light. Hence, the Sun and stars glow because the matter present in them exists predominantly in the **plasma state**. Therefore, the statement is correct.

Q 92. Ans. D

Explanation:

- **Statement 1 is correct:** **phosphorus** is an essential element present in important biological molecules such as **ATP (adenosine triphosphate)**, **DNA**, **RNA**, and **phospholipids**. The phosphate groups in ATP store and transfer energy required for cellular activities, while the sugar-phosphate backbone forms the structural framework of DNA and RNA. Therefore, phosphorus is indispensable for both **energy transfer** and the **structure of genetic material**, making the statement correct.
- **Statement 2 is correct:** **sodium (Na⁺)** and **potassium (K⁺)** ions play a vital role in the **generation and transmission of nerve impulses**. The movement of these ions across the nerve cell membrane creates electrical changes known as **action potentials**, which enable the conduction of nerve impulses. Thus, sodium and potassium are essential for the normal functioning of the nervous system, making the statement correct.
- **Statement 3 is correct:** **chloride ions (Cl⁻)** are required for the formation of **hydrochloric acid (HCl)** in the stomach. Hydrochloric acid provides the acidic medium necessary for the activation of **pepsin** from pepsinogen and facilitates the digestion of proteins. It also helps destroy many microorganisms present in food. Therefore, chlorine plays an important role in the **digestion process**, making the statement correct.

Q 93. Ans. B

Explanation:

- **Statement 1 is incorrect:** **isotopes** are atoms of the same element having the same **atomic number** (same number of protons) but **different mass numbers** because they contain **different numbers of neutrons**. Since the atomic number remains unchanged, isotopes exhibit similar chemical properties. Therefore, the statement is incorrect because it reverses the definition of isotopes.

- **Statement 2 is correct:** Isobars are atoms of different elements that have the same mass number but different atomic numbers. Because their atomic numbers differ, they contain different numbers of protons and exhibit different chemical properties, even though their mass numbers are identical. Therefore, the statement is correct.
- **Statement 3 is correct:** isoelectronic species are atoms or ions that possess the same number of electrons and hence have similar electronic configurations. For example, O^{2-} , F^- , Na^+ , Mg^{2+} , and Al^{3+} each contain 10 electrons and are therefore isoelectronic. Hence, the statement is correct.

Q 94. Ans. B

Explanation:

- **Water gas** is a gaseous mixture consisting mainly of carbon monoxide (CO) and hydrogen (H_2). It is produced by passing steam over red-hot carbon (coke) at high temperature. Water gas is used as an industrial fuel and as a source of hydrogen in various chemical processes. Therefore, the mixture of carbon monoxide and hydrogen is known as water gas.

Q 95. Ans. C

Explanation:

- **Borax** ($Na_2B_4O_7 \cdot 10H_2O$) is an important boron compound used in household and industrial applications. One of its key uses is in water softening, especially in reducing the effects of hard water.
- **Turpentine oil** is a volatile organic solvent obtained from the distillation of the resin of pine trees. It is widely used as a paint thinner because it can dissolve oil-based paints and varnishes.
- **DDT (Dichloro-Diphenyl-Trichloroethane)** is a synthetic chemical compound used as a powerful insecticide. It was widely used to control mosquitoes and agricultural pests. DDT works by affecting the nervous system of insects, leading to their death.
- **Epsom salt (magnesium sulphate, $MgSO_4 \cdot 7H_2O$)** is used medically as a laxative. It helps in stimulating bowel movement by drawing water into the intestines, thereby softening stools and aiding in the emptying of the bowels.

Q 96. Ans. C

Explanation:

- **Statement:** Decomposers mineralise dead organic matter.

- **Conclusion I:** decomposers such as bacteria and fungi break down dead plants and animals into simpler substances. During this process, called **decomposition and mineralisation**, complex organic matter is converted into simpler inorganic substances like carbon dioxide, water, nitrates, and phosphates. These nutrients are released back into the soil and environment, making them available again for plants. Therefore, decomposers play an important role in **recycling nutrients in the ecosystem**, so **Conclusion I is correct**.
- **Conclusion II:** Since mineralisation specifically refers to the conversion of organic matter into **inorganic nutrients**, decomposers actively release these inorganic substances into the environment during decomposition. These released nutrients mix with soil and water and are reused by producers, maintaining nutrient cycles in nature. Therefore, **Conclusion II is also correct**.

Q 97. Ans. B

Explanation:

- **Polyester (Terylene)** is a synthetic polymer formed by the condensation polymerisation of:
 - ✓ Ethylene glycol (a diol)
 - ✓ Terephthalic acid (a dicarboxylic acid)
- When these two monomers react, they form long-chain polymers called **polyesters**, with the elimination of water molecules. The most common polyester is **Terylene (also called Dacron)**, widely used in making clothes, curtains, and PET bottles.

Q 98. Ans. D

Explanation:

- **Pair 1 is correct:** a sol is a type of colloid in which the dispersed phase is solid and the dispersion medium is liquid. This means fine solid particles are uniformly distributed in a liquid medium without settling down. Common examples include ink, paint, and muddy water. These systems appear homogeneous to the naked eye but are actually heterogeneous at the microscopic level.
- **Pair 2 is correct:** gel as a colloidal system in which the dispersed phase is liquid, and the dispersion medium is solid. In this type of system, liquid is trapped within a solid network, giving a semi-solid structure. Examples include jelly, cheese, and butter. The liquid is immobilised inside the solid framework, which prevents it from flowing freely.
- **Pair 3 is correct:** an aerosol is a colloidal system in which either a solid or a liquid is dispersed in a

gas. This means the dispersion medium is always a gas, while the dispersed phase may be either solid or liquid. Examples include **fog (liquid in gas)**, **smoke (solid in gas)**, and **mist**. These systems are commonly seen in the atmosphere.

- **Pair 4 is correct:** foam as a colloidal system in which the **dispersed phase is gas** and the **dispersion medium is liquid**. In this system, gas bubbles are trapped within a liquid medium, forming a frothy structure. Common examples include **soap bubbles** and **shaving foam**. The gas remains dispersed throughout the liquid, giving the characteristic foamy appearance.

Q 99. Ans. C

Explanation:

- The **species** is considered the **fundamental and most specific unit of biological classification**. A species is defined as a group of organisms that are **similar in structure, function, and genetics**, and are capable of **interbreeding to produce fertile offspring**. It is the lowest and most basic rank in the taxonomic hierarchy. All higher taxonomic categories, such as genus, family, order, and phylum are made up of one or more species.

Q 100. Ans. C

Explanation:

- **Photosynthesis** in the chloroplast occurs in two main stages:
 - ✓ **Light reaction** → occurs in the **thylakoid membranes (grana)**
 - ✓ **Dark reaction (Calvin cycle)** → occurs in the **stroma of the chloroplast**
- The **stroma** is the fluid-filled matrix surrounding the grana inside the chloroplast. It contains enzymes required for the **Calvin cycle**, where **carbon dioxide is fixed into carbohydrates (glucose formation pathway)**. Since the dark reaction is enzyme-driven and does not directly require light, it takes place in the **stroma**, making it the actual site of the dark reaction.

Q 101. Ans. A

Explanation:

- **Mendel's laws of inheritance** were derived from experiments on the **pea plant (Pisum sativum)**, which reproduces by **sexual reproduction**. Sexual reproduction involves the formation of **gametes and their fusion (fertilisation)**, which allows segregation and recombination of alleles.

- Mendel studied patterns of inheritance such as **dominance, segregation, and independent assortment**, all of which become observable only when genes are passed through **gametes formed in sexual reproduction**. Therefore, Mendel's principles are fundamentally based on **sexual reproduction**.

Q 102. Ans. B

Explanation:

- **Ginger is a modified underground stem called a rhizome**. It grows horizontally under the soil and stores food. It has nodes, internodes, and buds, which are characteristics of a stem. Hence, ginger is a stem modification.

Q 103. Ans. B

Explanation:

- After **fertilisation** in a flowering plant, the **ovary of the flower develops into the fruit**. The ovary wall becomes the **fruit wall (pericarp)**, while the ovules inside the ovary develop into **seeds**. This transformation ensures the protection and dispersal of seeds. Thus, the fruit is essentially the **mature and ripened ovary**.
- The **ovule develops into the seed after fertilisation**, not the fruit. It contains the egg cell and is fertilised to form the embryo.

Q 104. Ans. A

Explanation:

- **Statement 1 is correct:** Athlete's foot (Tinea pedis) is a **fungal infection** that is **highly contagious**. It spreads through direct contact or indirectly via **contaminated objects such as towels, floors, socks, and shoes**. Warm and moist environments promote its transmission. Hence, this statement is correct.
- **Statement 2 is correct:** Fungal skin infections like Athlete's foot cause **itching, redness, and scaly white rashes**, particularly between the toes. The infection leads to irritation and discomfort due to fungal growth on the skin surface. Therefore, this statement is correct.
- **Statement 3 is incorrect:** Athlete's foot is caused by **fungi (dermatophytes)** such as Trichophyton. It is **not caused by helminths** (which are parasitic worms like roundworms and tapeworms). Helminths cause intestinal or tissue infections, not skin fungal infections. Hence, this statement is incorrect.

Q 105. Ans. D

Explanation:

- **Starfish** belong to the phylum **Echinodermata**, which includes exclusively marine organisms with a spiny skin. Members of this group show **radial symmetry in the adult stage** and possess a unique water vascular system used for movement and feeding. Starfish are commonly found on sea beds and exhibit slow movement using tube feet. Therefore, starfish is correctly classified under Echinodermata.
- **Whale** under the class **Mammalia**, even though it lives in water. Whales are **warm-blooded vertebrates**, breathe through lungs, and give birth to young ones instead of laying eggs. They also feed their young ones with milk produced by mammary glands, which is a defining feature of mammals. Hence, the whale is correctly placed under Mammalia.
- The **jellyfish** belongs to the phylum **Cnidaria (also called Coelenterata)**. These organisms are mostly marine and exhibit **radial symmetry**. They possess special stinging cells called **cnidoblasts**, which help in capturing prey and defence. Jellyfish have a simple body organisation and a sac-like body structure.
- **Silverfish** under the class **Insecta** within the phylum **Arthropoda**. It has a **segmented body divided into head, thorax, and abdomen**, and possesses jointed legs and an exoskeleton made of chitin. Silverfish are wingless insects commonly found in damp places and feed on starch-containing materials.

Q 106. Ans. B

Explanation:

- **Pair 1 is correct: chloroplasts** are green plastids containing chlorophyll. Their main function is **photosynthesis**, where light energy is converted into chemical energy by producing glucose from carbon dioxide and water. Chloroplasts are the primary sites of food production in green plants.
- **Pair 2 is correct: chromoplasts are coloured plastids** found in plant cells that are responsible for giving **distinct colours to flowers, fruits, and some leaves**. These colours arise due to the presence of **pigments such as carotenoids**, which include yellow, orange, and red pigments.
- **Pair 3 is incorrect: Leucoplasts are colourless plastids** found mainly in **non-green parts of plants**, such as **roots, seeds, tubers, and underground stems**. Since they lack pigments like chlorophyll

or carotenoids, they do not participate in photosynthesis or provide colouration. Their main function is the **storage of food materials**, making them essential for plant survival during non-photosynthetic conditions.

- **Pair 4 is incorrect: elaioplasts are a specialised type of leucoplast** that are specifically involved in the **storage of lipids (oils and fats)**. Since leucoplasts are colourless plastids primarily associated with storage, elaioplasts represent the lipid-storing category within this group.

Q 107. Ans. D

Explanation:

- **magnetic flux density (B)** is a physical quantity that represents the strength of a magnetic field at a point in space. It is defined in terms of the force experienced by a current-carrying conductor placed in a magnetic field. Magnetic flux density can be understood as the force acting per unit current and per unit length of the conductor. The SI unit derived from this definition is **Newton per ampere per metre ($\text{N A}^{-1} \text{m}^{-1}$)**, which is named **Tesla (T)**.

Q 108. Ans. A

Explanation:

- The **minimum velocity required for a satellite to revolve around the Earth in a stable circular orbit** is called **orbital velocity**. When an object is given this velocity tangentially to the Earth's surface at a suitable height, it begins to continuously fall towards the Earth due to gravity, but because of its high horizontal speed, it keeps missing the Earth and thus remains in a circular orbit. This balance between gravitational force (providing centripetal force) and the satellite's inertia is what maintains orbital motion.

Q 109. Ans. D

Explanation:

- A **cohesive force** is the force of attraction between molecules of the **same substance**. This force is responsible for holding the particles of a solid together in a rigid structure. In **solids**, molecules are very closely packed, so cohesive forces are **maximum**, giving solids a definite shape and volume. In **liquids**, cohesive forces are moderate, allowing flow. In **gases**, molecules are far apart, so the intermolecular attraction becomes **negligible**. Therefore, cohesive force is strongest in solids and weakest (almost negligible) in gases.

Q 110. Ans. B**Explanation:**

- **Statement 1 is incorrect:** the refractive index is defined as the ratio of the speed of light in vacuum to the speed of light in a given medium. Since it is a ratio of two scalar quantities and has no associated direction, the refractive index is a **scalar quantity**, not a vector. Therefore, the statement that refractive index is a vector is incorrect.
- **Statement 2 is correct:** electric current is defined as the rate of flow of electric charge. Although current has a direction of flow in a circuit, it does not follow the rules of vector addition like true vector quantities (such as displacement or force). Instead, currents add algebraically depending on direction. Hence, electric current is treated as a **scalar quantity**, making the second statement correct.

Q 111. Ans. C**Explanation:**

- **Statement 1 is correct:** the speed of sound in a gas depends on properties of the medium such as temperature, humidity, and the nature of the gas, but not on pressure when temperature is kept constant. This is because, at constant temperature, an increase in pressure is accompanied by a proportional increase in density, and their ratio remains unchanged, so the speed of sound does not change.
- **Statement 2 is correct:** the speed of sound in air increases with an increase in humidity. This is because moist air is less dense than dry air due to the lower molecular mass of water vapour compared to dry air components. As a result, sound travels faster in humid air.

Q 112. Ans. A**Explanation:**

- **Statement 1 is correct:** light travels in a straight line when it moves through a **homogeneous medium** (a medium having uniform optical properties). This is known as the **rectilinear propagation of light**. Hence, the first statement is correct.
- **Statement 2 is incorrect:** refraction of light occurs due to a change in the speed of light when it passes from one medium to another, not due to a change in frequency. The **frequency of light remains constant** during refraction, while its speed and wavelength change. Since the statement incorrectly attributes refraction to a change in frequency, the second statement is **incorrect**.

Q 113. Ans. D**Explanation:**

- **Statement:** Barrier methods prevent the meeting of sperm and ovum.
- **Conclusion I:** Barrier methods such as condoms and diaphragms work by physically blocking the sperm from reaching the ovum, thereby preventing fertilisation. Since fertilisation is essential for pregnancy, this mechanism helps in avoiding unwanted pregnancies. Therefore, the given statement implies that barrier methods are effective contraceptive techniques, making **Conclusion I correct**.
- **Conclusion II:** Barrier methods act only as physical barriers and do not interfere with hormonal control or the process of ovulation in the female body. Ovulation, which is the release of an egg from the ovary, continues normally even when barrier methods are used. Hence, the claim that these methods permanently inhibit ovulation is incorrect, making **Conclusion II wrong**.

Q 114. Ans. D**Explanation:**

- **Indian Railways** has established the **CHIRAG (Centre for HR Information Systems, Research and Analytics in Governance)** Centre at Moula-Ali, Hyderabad, adjacent to the **Zonal Railway Training Institute**, to modernise its human resource (HR) governance. Designed to strengthen workforce management for one of the world's largest employers, the centre leverages **Artificial Intelligence (AI), Machine Learning (ML), and Predictive Analytics** to improve HR operations and support data-driven decision-making. CHIRAG will also provide **role-based Human Resource Management System (HRMS) training**, develop **master trainers**, and serve as a **think tank** for shaping future HR policies and governance practices within Indian Railways.

Q 115. Ans. B**Explanation:**

- **Coal (1) — sedimentary:** Coal is a classic example of an organic sedimentary deposit, formed in horizontal beds/strata over geological time.
- **Bauxite (2) — NOT sedimentary (residual/weathering origin):** Despite often being found within geologically "tertiary" formations, bauxite itself forms through intense chemical weathering

(laterization) of aluminium-silicate rocks, leaving a residual deposit that is explicitly distinguished from the sedimentary mode of formation in the standard curriculum.

- **Clay (3) — sedimentary:** Clay is a classic sedimentary material/rock, formed via deposition and accumulation of fine weathered particles.

Q 116. Ans. C

Explanation:

- **Statement:** The placenta facilitates the exchange of nutrients, gases and metabolic wastes between the mother and the foetus.
- **Conclusion I:** The **placenta** is a special organ formed in the uterus that connects the developing foetus to the maternal body. It acts as an interface for the **exchange of nutrients, oxygen, carbon dioxide, and metabolic wastes** between the mother and the foetus. Since this exchange is essential for growth, nutrition, and survival of the foetus, the placenta directly **supports foetal development**, making the **conclusion I correct**.
- **Conclusion II:** Although exchange of substances occurs through the placenta, there is **no direct mixing of maternal and foetal blood**. The two blood supplies remain separate, and exchange happens through a thin placental membrane by diffusion and other transport mechanisms. Therefore, **Conclusion II is also correct**

Q 117. Ans. C

Explanation:

- **Red Balloon Aerospace** successfully launched India's first indigenous stratospheric super-pressure balloon under Mission SANA. Developed by the Vijayawada-based startup, the VISTA platform reached an altitude of approximately 25 km in the stratosphere. With this achievement, India joined the United States, France, Japan, and China as one of the few countries possessing indigenous stratospheric hydrogen super-pressure balloon technology.
- Unlike conventional weather balloons, which expand and burst as they ascend due to decreasing atmospheric pressure, **super-pressure balloons maintain a stable, pressurised shape throughout the flight**. This design enables them to remain at a nearly constant altitude for **weeks or even months**, making them well-suited for long-duration atmospheric research, Earth observation, and communication-related applications.

Q 118. Ans. B

Explanation:

- **National Technology Day 2026** was celebrated across India on **11 May 2026** with the theme **"Responsible Innovation for Inclusive Growth."** The celebration emphasised India's indigenous technological achievements in areas such as **artificial intelligence, space exploration, and digital public infrastructure**. The flagship event, **Vigyan Tech 2026 (विज्ञानTech 2026)**, was held at the **National Institute of Immunology (NII), New Delhi**. The exhibition featured **patent-driven technologies** developed by various government science departments, with a focus on promoting innovation, supporting startups, and enabling the growth of small and medium enterprises (SMEs).

Q 119. Ans. B

Explanation:

- **Statement:** Natural selection acts upon heritable variations present in a population.
- **Conclusion I:** Natural selection does not create variations; rather, it acts on the **already existing heritable variations** present in a population. Variations arise due to mutations and genetic recombination, and natural selection only filters them based on environmental advantage. Hence, the idea that natural selection creates useful variations according to the needs of organisms is incorrect, making **Conclusion I wrong**.
- **Conclusion II:** Individuals possessing **advantageous heritable traits** have a better chance of survival and reproduction. These individuals are more likely to leave behind more offspring, thereby passing on beneficial traits to the next generation. This is the core idea of natural selection. Therefore, **Conclusion II is correct**.

Q 120. Ans. C

Explanation:

- **Statement 1 is correct:** **Prajna** is an indigenous AI-enabled satellite intelligence system handed over by DRDO to the **Ministry of Home Affairs (MHA)**. It supports **border management** by analysing satellite imagery to help security forces monitor sensitive regions, detect suspicious activities, and strengthen surveillance capabilities.
- **Statement 2 is incorrect:** **Prajna** was developed by the Defence Research and Development Organisation's (DRDO) Centre for Artificial

Intelligence and Robotics (CAIR), not by the Indian Space Research Organisation (ISRO).

- **Statement 3 is correct:** The platform automatically processes large volumes of satellite imagery using advanced machine learning analytics. It can identify anomalies and suspicious movements in real time, making it an important tool for border security, surveillance, and counter-terrorism operations.

Q 121. Ans. A

Explanation:

- According to the India State of Forest Report (ISFR) published by the Forest Survey of India (FSI), “tree cover” refers to patches of trees and isolated trees located outside the Recorded Forest Area (RFA). These include trees growing on farmlands, along canals and roads, in urban areas, institutional campuses, village commons, and other non-forest lands. Specifically, tree cover comprises tree patches of less than one hectare and scattered trees that are not included under forest cover.

Q 122. Ans. D

Explanation:

- **Quinary activities:** the highest tier, involving top-level decision-making, policy formulation, and the creation/interpretation of new ideas and technologies — the domain of senior executives, top government officials, research scientists, and financial/legal consultants. Workers here are termed “gold-collar” professionals, reflecting their specialised, highly compensated, and decision-critical roles.
- **Additional Facts:**
 - ✓ **Primary activities** (agriculture, mining, fishing, forestry): workers are called red-collar workers, given the outdoor nature of the work.
 - ✓ **Secondary activities** (manufacturing, construction, processing): workers are called blue-collar workers.
 - ✓ **Tertiary activities** (trade, transport, general services): workers are called white-collar workers.
 - ✓ **Quaternary activities** (knowledge-sector services like IT, research support, consultancy, data analysis): these form a specialised subdivision of the tertiary sector.

Q 123. Ans. D

Explanation:

- **Shifting cultivation** (slash-and-burn agriculture) is practised widely across the tropics under different local names. All four pairs given are correctly

matched, per standard geography references:

- ✓ **Tamrai — Thailand:** correct, the traditional local name for shifting cultivation in Thailand.
- ✓ **Caingin (also spelt Kaingin) — Philippines:** correct, widely documented as the Philippine term for slash-and-burn cultivation, still practised in upland regions like Leyte Island.
- ✓ **Humah — Java:** correct, the local name specific to Java (Indonesia), distinct from “Ladang,” which is the broader term used in Indonesia/Malaysia generally.
- ✓ **Chena — Sri Lanka:** correct, the traditional dry-zone shifting cultivation practice of Sri Lanka.

Q 124. Ans. D

Explanation:

- **Statement:** Energy transfer in a food chain, only about 10% of the energy is passed from one trophic level to the next, while the remaining 90% is lost as heat through respiration and other life processes.
- **Conclusion I:** Only about 10% of energy available at one trophic level is transferred to the next trophic level. The remaining energy is lost mainly as heat through metabolic activities. Because of this continuous loss at each step, the total energy decreases progressively along the food chain. Therefore, Conclusion I is correct.
- **Conclusion II:** energy flow in an ecosystem is unidirectional, moving from producers to consumers and then to decomposers, and it is not returned back to previous trophic levels. Hence, the statement that energy flow is bidirectional is incorrect. Therefore, Conclusion II is wrong.

Q 125. Ans. D

Explanation:

- **Kandla Port — Gujarat**
 - ✓ Also known as Deendayal Port, it is a major tidal port located on the Gulf of Kutch and is one of India’s leading ports for handling petroleum products and dry cargo.
- **Ennore Port — Tamil Nadu**
 - ✓ Officially known as Kamarajar Port, it is located near Chennai and is India’s first corporatised major port, handling coal, automobiles, LNG, and container cargo.
- **Paradip Port — Odisha**
 - ✓ Situated on the Bay of Bengal, Paradip is a major natural deep-water port and a key gateway for the export of minerals, coal, and petroleum products.

- Mormugao Port — Goa
 - ✓ Located on the Zuari Estuary, Mormugao is one of India's oldest natural harbours and is particularly important for the export of iron ore.

Q 126. Ans. C

Explanation:

- **Carrot:** This is primarily a temperate, cool-season crop and is not suited for the constant high temperatures and heavy rainfall of equatorial regions.
- **Sugarcane, Coffee, and Tea:** These are major plantation and cash crops that thrive in hot, wet equatorial and tropical climates. Geographical classifications explicitly identify sugarcane (sugar), coffee, and tea alongside rubber, cocoa, and coconuts as crops actively cultivated in equatorial regions due to the favourable uniform temperature and consistent heavy rainfall.

Q 127. Ans. B

Explanation:

- An anticyclone is a high-pressure system in which air descends (subsides) towards the Earth's surface. As the air sinks, it warms adiabatically, reducing relative humidity and inhibiting cloud formation. Consequently, anticyclones are typically associated with stable weather, clear skies, light winds, and dry conditions.

Q 128. Ans. C

Explanation:

- Karanprayag is one of the Panch Prayag (five sacred confluences) of the Alaknanda River in Uttarakhand. It is formed by the confluence of:
 - ✓ Alaknanda River
 - ✓ Pindar River
- The five Prayags, in downstream order, are:
 - ✓ Vishnuprayag – Alaknanda + Dhauliganga
 - ✓ Nandprayag – Alaknanda + Nandakini
 - ✓ Karnaprayag – Alaknanda + Pindar
 - ✓ Rudraprayag – Alaknanda + Mandakini
 - ✓ Devprayag – Alaknanda + Bhagirathi (where the river thereafter is known as the Ganga)

Q 129. Ans. D

Explanation:

- Continental Drift Theory
 - ✓ **Proponent:** Alfred Wegener (1912)
 - ✓ All continents were once joined together as a single supercontinent called Pangaea, surrounded by a single ocean, Panthalassa.

- ✓ Evidence cited includes jigsaw-fit of coastlines (Africa–South America), similar rock formations and fossils (e.g., *Glossopteris*) across continents, and matching glacial deposits.

● Convection Current Theory

- ✓ **Proponent:** Arthur Holmes (1930s)
 - Proposed that thermal convection currents within the Earth's mantle (driven by radioactive heat) provide the driving force for continental movement — addressing the gap left by Wegener.

● Plate Tectonic Theory

- ✓ **Proponents:** Developed collectively (McKenzie, Parker, Morgan) in the late 1960s, building on Wegener's ideas and sea-floor spreading (Harry Hess, 1960s).
 - The Earth's lithosphere is broken into several rigid plates that float on the semi-fluid asthenosphere and move due to mantle convection, causing three types of plate boundaries: divergent (constructive), convergent (destructive), and transform (conservative).

● Central Place Theory

- ✓ **Proponent:** Walter Christaller (1933)
- ✓ Explains the number, size, and distribution of human settlements (urban centres) in a system, based on the concept that settlements function as “central places” providing goods and services to surrounding areas, arranged in a hexagonal hierarchy.

Q 130. Ans. A

Explanation:

- Ocean currents are driven by a combination of wind, temperature and salinity differences, Earth's rotation (Coriolis effect), and the configuration of continents.
 - ✓ **Upwelling** – Upwelling is the upward movement of cold, nutrient-rich deep water to the ocean surface. It is closely linked to the movement of ocean currents and is influenced by prevailing winds and the Coriolis effect.
 - ✓ **Thermohaline circulation** – Also known as the global ocean conveyor belt, it is driven by differences in temperature (thermo) and salinity (haline). It plays a major role in controlling deep ocean currents and redistributing heat across the globe.
 - ✓ **Tsunamis** – Tsunamis are large seismic sea waves generated by underwater earthquakes,

volcanic eruptions, or landslides. They are short-lived wave phenomena and do not influence the long-term movement or circulation of ocean currents.

Q 131. Ans. B

Explanation:

- The Government of India Act of 1919, known as the Montagu-Chelmsford Reforms, introduced dyarchy in the provinces by dividing the provincial subjects into two categories, transferred and reserved; the transferred subjects were administered by the Governor with the aid of ministers responsible to the legislative council, while the reserved subjects were administered by the Governor and his executive council without such responsibility.
- It was the Government of India Act of 1935, and not the Act of 1919, that introduced provincial autonomy in place of dyarchy in the provinces; the 1919 Act introduced dyarchy, which the 1935 Act later abolished by making the provinces autonomous units.
- The Act of 1919 also relaxed the central control over the provinces by demarcating central and provincial subjects, introduced bicameralism and direct elections at the centre, required three of the six members of the Viceroy's executive council to be Indians, extended communal representation to further communities, and provided for the establishment of a public service commission and a periodic statutory commission to examine its working, which materialised as the Simon Commission.

Q 132. Ans. C

Explanation:

- The Indian Independence Act of 1947 ended the British paramountcy over the Indian princely states and the treaty relations with the tribal areas, leaving the princely states free to join either of the two new dominions of India and Pakistan or to remain independent.
- The Act abolished the office of the Secretary of State for India and transferred his functions to the Secretary of State for Commonwealth Affairs, since the apparatus of imperial control over India was no longer required once independence was granted.
- The Act declared India and Pakistan to be independent dominions from 15 August 1947, empowered the constituent assemblies of the two dominions to frame and adopt any constitution and to repeal any act of the British Parliament,

including the Independence Act itself, and made the Governor General and the provincial Governors constitutional heads acting on the advice of the respective councils of ministers. It was based on the Mountbatten Plan of 3 June 1947.

Q 133. Ans. A

Explanation:

- During a national emergency, the otherwise federal Constitution can be converted into a unitary one, since the Centre becomes entitled to give directions to the states on any matter, Parliament can legislate on subjects in the State List, and the distribution of revenues can be altered, so that the states come under the complete control of the Centre without a formal suspension of the Constitution.
- The emergency provisions are not borrowed from the United States; the concept of the suspension of fundamental rights during an emergency is drawn largely from the Weimar Constitution of Germany, while the broader scheme of emergency powers reflects the Government of India Act of 1935. The American Constitution, by contrast, does not contain such elaborate emergency provisions.
- This capacity to transform the federal structure into a unitary one in times of crisis is regarded as one of the salient features and unique characteristics of the Constitution, which the framers justified as necessary for safeguarding the sovereignty, unity, integrity, and security of the country, the democratic political system, and the Constitution itself.

Q 134. Ans. A

Explanation:

- **Statement 1 is correct:** The expression martial law is not defined anywhere in the Constitution and is implicit in Article 34; it is borrowed from English common law and signifies the suspension of ordinary law and the temporary government of an area by military authority to restore order during a breakdown of law and order.
- **Statement 2 is incorrect:** A national emergency does not automatically suspend the operation of all fundamental rights; under Article 358 the six freedoms under Article 19 are automatically suspended only when the emergency is declared on the ground of war or external aggression, and under Article 359 the enforcement of other specified rights can be suspended by a presidential order, but the rights under Articles 20 and 21 can never be suspended.

- Martial law differs from a national emergency in that it affects only the fundamental rights in a localised area, suspends the ordinary courts and the government of that area, and is imposed only in the case of a breakdown of law and order, whereas a national emergency affects the fundamental rights, the centre state relations, the distribution of revenues, the legislative powers, and the term of the Lok Sabha and the state assemblies.

Q 135. Ans. D

Explanation:

- **Article 47 is unique because its text contains two distinct directives that fall under two different ideological classifications:**
 - ✓ **Socialist Principle:** The first part of the article states, "The State shall regard the raising of the level of nutrition and the standard of living of its people and the improvement of public health as among its primary duties." This aligns with the socialist goal of a welfare state.
 - ✓ **Gandhian Principle:** The second part states, "...and, in particular, the State shall endeavour to bring about prohibition of the consumption except for medicinal purposes of intoxicating drinks and of drugs which are injurious to health." This aligns directly with Gandhi's ideology regarding temperance and prohibition.

Q 136. Ans. A

Explanation:

- **Statement 1 is correct:** The Prime Minister recommends to the President the persons who can be appointed as ministers, and the President appoints only those persons as ministers who are recommended by the Prime Minister, so the composition of the Council of Ministers is effectively determined by the Prime Minister.
- **Statement 2 is correct:** The Prime Minister allocates and reshuffles the portfolios among the ministers, can ask a minister to resign or advise the President to dismiss a minister in case of a difference of opinion, and presides over the meetings of the Council of Ministers, thereby guiding, directing, controlling, and coordinating the activities of all the ministers.
- **Statement 3 is incorrect:** The resignation or death of the Prime Minister does affect the Council of Ministers, because it automatically dissolves the entire Council and brings about its collapse, creating a vacuum, whereas the resignation or death of any other minister merely creates a vacancy that the Prime

Minister may or may not choose to fill. The Prime Minister is thus the keystone of the cabinet arch.

Q 137. Ans. D

Explanation:

- Public interest litigation can be used to enforce the rights of a determinate class of persons, such as prisoners, bonded labourers, or inmates of protective homes, who by reason of poverty, disability, or a socially or economically disadvantaged position are unable to approach the court for relief themselves.
- Public interest litigation is not an adversarial form of litigation between two contesting parties; it is essentially a collaborative and non-adversarial proceeding in which the petitioner, the State, and the court work together to find a solution to a problem affecting the public interest, with the court often appointing commissions to gather facts and monitoring the implementation of its directions.
- The scope of public interest litigation extends to the protection and enforcement of the fundamental rights of those who cannot seek relief, the vindication of the rule of law, and the redress of public injury arising from the breach of a public duty. The courts have, however, framed guidelines to ensure that it is invoked for genuine public causes and not for private, political, or publicity interests.

Q 138. Ans. A

Explanation:

- **Statement 1 is correct:** Under Article 164, the Chief Minister is appointed by the Governor, and the other ministers are appointed by the Governor on the advice of the Chief Minister, so that the Governor's formal power of appointment is, in practice, exercised in accordance with the choices of the Chief Minister.
- **Statement 2 is incorrect:** The total number of ministers in a state cannot exceed fifteen per cent of the total strength of the legislative assembly, not twelve per cent, subject to a minimum of twelve ministers, a limit introduced by the 91st Amendment Act of 2003 to curb the practice of forming large ministries for political accommodation.
- The Council of Ministers in a state is collectively responsible to the legislative assembly, and the ministers hold office during the pleasure of the Governor. The same amendment also provided that a member disqualified on the ground of defection shall be disqualified from being appointed a minister until he is re-elected, a measure intended to strengthen the anti-defection regime.

Q 139. Ans. A**Explanation:**

- Under Article 156, the Governor holds office during the pleasure of the President, which means that the Governor has no fixed security of tenure and can be removed by the President at any time, though the normal term of office is five years from the date of entering upon the office.
- A person must have completed thirty-five years of age, not thirty, to be eligible for appointment as Governor; the other qualifications under Article 157 are that the person must be a citizen of India, and by convention, the person should be an outsider not belonging to the state and should be consulted along with the Chief Minister before appointment.
- The Governor is appointed by the President by warrant under his hand and seal and is, in effect a nominee of the central government. The Governor must not be a member of either House of Parliament or of a state legislature, must not hold any other office of profit, and is entitled to emoluments and the use of an official residence; the same person can be appointed Governor of two or more states.

Q 140. Ans. A**Explanation:**

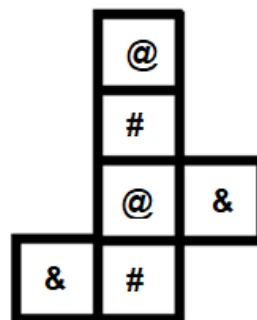
- **Statement 1 is correct:** Article 343 declares Hindi in the Devanagari script to be the official language of the Union, with the form of numerals to be the international form of Indian numerals, marking Hindi as the principal official language for the purposes of the central government.
- **Statement 2 is incorrect:** Article 343 did not abolish the use of English after fifteen years; it provided that English would continue to be used for the official purposes of the Union for a period of fifteen years from the commencement of the Constitution, and the Official Languages Act of 1963 later provided for the indefinite continuation of English alongside Hindi, so that English was not abolished.
- Article 344 provides for an Official Language Commission and a Committee of Parliament on Official Language, Article 345 allows a state legislature to adopt one or more languages in use in the state as its official language, Article 348 provides for the use of English in the higher judiciary and in the texts of laws, and Article 351 directs the Union to promote the spread of Hindi.

Q 141. Ans. A**Explanation**

- The number of odd days in a normal year is 1, and the number of odd days in leap year is 2.
- There are 4 non-leap years (2021, 2022, 2023, 2025) and 2 leap years 2020 and 2024 from January 6, 2020 to January 6, 2026.
- So, number of odd days will be,
 - ✓ From January 6, 2020 to January 5, 2021: 2
 - ✓ From January 6, 2021 to January 5, 2022: 1
 - ✓ From January 6, 2022 to January 5, 2023: 1
 - ✓ From January 6, 2023 to January 5, 2024: 1
 - ✓ From January 6, 2024 to January 5, 2025: 2
 - ✓ From January 6, 2025 to January 5, 2026: 1
- So, total number of odd days are 8 which is divided by 7 to get the final number of odd days i.e., 1
- So, on January 6, 2026, it will be Tuesday (As Monday +1 odd day = Tuesday)

Q 142. Ans. C**Explanation:**

- The logic followed here is:
 - ✓ $2^3 + 1 = 9$
 - ✓ $3^3 + 2 = 29$
 - ✓ $4^3 + 3 = 67$
 - ✓ $5^3 + 4 = 129$
 - ✓ $6^3 + 5 = 221$
 - ✓ $7^3 + 6 = 349$

Q 143. Ans. C**Explanation:**

- A is opposite to C
- B is opposite to F.
- D is opposite to E

Q 144. Ans. A**Explanation:**

- Since,

W	R	I	T	E
+2↓	+2↓	+2↓	+2↓	+2↓
Y	T	K	V	G

- In the same way,

R	E	A	D
+2↓	+2↓	+2↓	+2↓
T	G	C	F

Q 145. Ans. C**Explanation:**

- The correct dictionary order will be:
- 5) Partial, 2) Participle, 1) Particle, 4) Particular, 3) Partisan

Q 146. Ans. B**Explanation:**

- Sum of the three prime numbers = 118
- If the sum of three prime numbers is even, then one of the numbers must be 2!
- (Because 2 is the only even prime number.)
- First number = 2
- Second number = p
- Third number = p + 42 (42 greater than the second number)
- $2 + p + (p + 42) = 118$
- $2 + 2p + 42 = 118$
- $2p + 44 = 118$
- $2p = 74$
- $p = 37$
- Largest number = $37 + 42 = 79$

Q 147. Ans. B**Explanation:**

- $3/8 = 0.375$, which is a terminating decimal and is also a rational number.
- Whereas,
- $2/7 = 0.285714...$ (Non-terminating repeating decimal)
- $\sqrt{5}$ = Is an irrational number.
- π = Is an irrational number.

Q 148. Ans. C**Explanation:**

- According to the question,
 - ✓ The two-digit prime numbers satisfying the above condition are: 13, 17, 31, 37, 71, 73, 79, 97
 - ✓ The sum of all such numbers:
 - $= 13 + 17 + 31 + 37 + 71 + 73 + 79 + 97$
 - ✓ $= 418$

Q 149. Ans. A**Explanation:**

- Since the number of rows and the number of students in each row are equal, the total number of students must be a perfect square.
- The nearest perfect square number greater than 950 is:
 - ✓ $31^2 = 961$
- $\therefore$ Required number of students = $961 - 950 = 11$

Q 150. Ans. B**Explanation:**

- Total saleable units produced per day = $2,400 \times 85\% = 2,040$ units
- Total demand for the product = 71,400 units.
 - ✓ Therefore, the time required to complete the order = $71,400/2,040$
 - $= 35$ days.

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